APPLIED CASEBOOK

Equity Valuation Casebook for MBA Students

Detailed Solutions

73 application-oriented cases spanning eleven sections of equity valuation, from foundational concepts to leveraged buyouts.

Each case pairs a realistic business scenario with a complete, step-by-step solution built for independent verification.

Designed for MBA students building advanced, decision-oriented valuation judgment beyond formula recall.

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Purpose of the Casebook

This casebook contains 73 application-oriented exercises in equity valuation, organized into eleven sections that progress from foundational concepts through dividend discount models, free cash flow valuation, terminal value, market multiples, enterprise value and financing effects, economic value added, and leveraged buyout and integrated valuation applications. Each case is framed as a realistic Indian business, investment, or advisory situation, requiring calculation, interpretation, and an investment or managerial decision — not mere formula recall.

Intended Audience

This casebook is intended for MBA students who already know the fundamentals of investment management, understand basic financial statements, are comfortable with basic mathematics and statistics, and know the time value of money and standard valuation formulae. It is designed to build an advanced, application-oriented understanding of equity valuation rather than to introduce first principles.

Learning Approach

Each case follows a consistent structure: a stated learning objective, a concise business situation (typically 75-100 words), two or three focused requirements, and a solution appropriate to the edition. Cases are designed to be used independently, in class discussion, as assignments, or for self-study, and generally do not require an earlier case to be solved first.

Organization of the Casebook

Section I: Valuation Foundations and Financial Statement Preparation (Cases 1-6)

Section II: Dividend Discount Models and Growth Opportunities (Cases 7-17)

Section III: FCFF, FCFE, and Explicit-Period Cash Flows (Cases 18-27)

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Symbols and Abbreviations

₹: Indian Rupee

EPS: Earnings Per Share

ROE: Return on Equity

Ke: Cost of Equity

WACC: Weighted Average Cost of Capital

g: Growth Rate

b: Retention Ratio

DDM: Dividend Discount Model

FCFF: Free Cash Flow to the Firm

FCFE: Free Cash Flow to Equity

EBIT: Earnings Before Interest and Tax

EBITDA: Earnings Before Interest, Tax, Depreciation and Amortisation

NOPAT: Net Operating Profit After Tax

PVGO: Present Value of Growth Opportunities

P/E: Price-to-Earnings Ratio

P/B: Price-to-Book Ratio

P/S: Price-to-Sales Ratio

P/CF: Price-to-Cash-Flow Ratio

PEG: P/E-to-Growth Ratio

EV: Enterprise Value

APV: Adjusted Present Value

EVA: Economic Value Added

MVA: Market Value Added

LBO: Leveraged Buyout

IRR: Internal Rate of Return

CMP: Current Market Price

cr: crore (1 crore = 10 million)

General Rounding Convention

Currency values are rounded to two decimal places; ratios are rounded to two decimal places; percentages are rounded to two decimal places. Minor differences between figures may arise because of rounding at intermediate steps.

Note on Classroom Use

The detailed solutions and teaching insights in this edition are intended to support in-class walkthroughs and grading. The 'Teaching Insight' accompanying most cases highlights a common student error or a natural classroom discussion point, and can be used to open or close discussion of the case.

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SECTION I

Valuation Foundations and Financial Statement Preparation

Case 1: Intrinsic Value versus Market Price

Learning Objective: Distinguish between market price and intrinsic value and convert the difference into an investment recommendation.

Case Situation

Suryodaya Auto Components Ltd, an NSE-listed manufacturer of forged auto-ancillary parts, currently trades at ₹410 per share. An equity research analyst tracking the stock has estimated its intrinsic value at ₹480 per share, based on the present value of the company's expected future free cash flows. The analyst's investment committee wants a clear recommendation before the next portfolio review, since stock has traded in a narrow range for several months despite the company reporting steady order-book growth.

Required

1. Calculate the percentage valuation gap between intrinsic value and market price.
2. Recommend whether the stock should be bought, held, or avoided.
3. Explain why market price need not immediately converge to intrinsic value.

Solution

A. Relevant Valuation Principle: Intrinsic value is the analyst's estimate of fundamental worth based on discounted future cash flows; market price is the price set by trading activity and need not equal intrinsic value at any given time.

B. Formula: Valuation gap (%) = (Intrinsic Value − Market Price) / Market Price × 100

C. Substitution and Working:

Valuation gap = (480 − 410) / 410 × 100 = 70 / 410 × 100 = 17.07%

D. Final Result: The stock is trading at a 17.07% discount to estimated intrinsic value.

E. Interpretation: A gap of this size is economically meaningful (well above typical bid-ask/estimation noise) and suggests the market may not yet be pricing in the order-book growth the analyst has factored into cash-flow projections. The gap by itself does not prove the market is wrong — it could also mean the analyst's assumptions are too optimistic.

F. Decision/Recommendation: With a margin of safety above 15%, and assuming the underlying assumptions are reasonable, the analyst can recommend Buy, with a target price aligned to the ₹480 intrinsic value estimate.

G. Teaching Insight: Emphasise that market price reflects the collective, often short-horizon view of many participants, while intrinsic value reflects one analyst's long-horizon fundamental estimate. Convergence, if it happens, depends on catalysts (results, re-rating, news flow) and can take time — a valuation gap is a hypothesis, not a guarantee of price movement. Common student error: treating any positive gap as an automatic 'buy' without asking how confident the cash-flow assumptions are.

Case 2: Equity Value versus Enterprise Value

Learning Objective: Understand the distinction between the value of operating assets, enterprise value, and the value attributable to equity shareholders.

Case Situation

Trident Logistics Ltd, a listed Indian logistics company, has an estimated operating-asset value (enterprise value) of ₹2,400 crore based on its core freight and warehousing business. The company carries total debt of ₹600 crore and holds excess cash of ₹150 crore beyond its operating needs, along with a small non-operating investment in an associate warehousing venture valued at ₹50 crore. The company has 100 crore shares outstanding. The CFO wants to understand how these components translate into value per share for equity holders.

Required

4. Calculate the enterprise value and the equity value of the company.
5. Calculate the equity value per share.
6. Explain why debt is deducted while excess cash and the non-operating investment are added.

Solution

A. Relevant Valuation Principle: Enterprise value represents the value of the firm's core operating assets, available to all capital providers. Equity value is what remains for shareholders after settling debt claims, plus any value not reflected in operations (excess cash, non-operating investments).

B. Formula: Equity Value = Enterprise Value − Debt + Excess Cash + Non-operating Investments

C. Substitution and Working:
Equity Value = 2,400 − 600 + 150 + 50 = ₹2,000 crore

Equity Value per share = 2,000 crore / 100 crore shares = ₹20.00

D. Final Result: Enterprise Value = ₹2,400 crore; Equity Value = ₹2,000 crore; Equity Value per share = ₹20.00.

E. Interpretation: Enterprise value reflects only the operating business. Debt holders have a prior claim on that value, so debt is deducted. Excess cash and the associate stake are assets that exist alongside — but are not part of — the operating business, so they are added back to arrive at what belongs to equity.

F. Decision/Recommendation: The per-share equity value of ₹20.00 can now be directly compared with the market price to assess relative value; the CFO can also use this bridge to explain to investors how much of firm value sits in operations versus non-operating assets.

G. Teaching Insight: A frequent student error is deducting gross debt without adding back cash (double-penalizing the firm) or ignoring non-operating investments entirely. Stress that 'enterprise value' already excludes non-operating items by definition — they must be added separately.

Case 3: Normalizing Reported Earnings

Learning Objective: Adjust reported profit for exceptional and non-recurring items before using earnings in valuation.

Case Situation

Kaveri Textiles Ltd, a family-managed textile company, reported a profit after tax of ₹85 crore for the year — well above analyst expectations. On closer review, the profit includes an exceptional gain of ₹18 crore from the sale of surplus factory land and a one-time insurance settlement gain of ₹7 crore following a fire at an old unit. The company has 10 crore shares outstanding. A junior analyst has drafted a valuation note using the reported EPS directly in a P/E-based valuation.

Required

7. Calculate the normalised PAT and normalised EPS.
8. Compare reported EPS with normalised EPS.
9. Explain why reported EPS should not be used directly in the P/E valuation.

Solution

A. Relevant Valuation Principle: Valuation multiples should be based on sustainable, recurring earnings; one-time gains inflate reported profit without indicating repeatable earning power.

B. Formula: Normalised PAT = Reported PAT − Exceptional/Non-recurring Gains

C. Substitution and Working:

Normalised PAT = 85 − 18 − 7 = ₹60 crore

Reported EPS = 85 / 10 = ₹8.50

Normalised EPS = 60 / 10 = ₹6.00

D. Final Result: Normalised PAT = ₹60 crore; Normalised EPS = ₹6.00 (versus reported EPS of ₹8.50).

E. Interpretation: Nearly 29% of reported profit (₹25 crore of ₹85 crore) comes from non-recurring items unrelated to the core textile business. Using reported EPS would overstate the company's sustainable earning power and produce an inflated P/E-based value.

F. Decision/Recommendation: The valuation note should be revised to use the normalised EPS of ₹6.00 as the basis for any P/E valuation; the exceptional gains should be disclosed separately as one-time items, not embedded in the earnings base.

G. Teaching Insight: A common error is normalising only for the more obvious item (land sale) while overlooking smaller exceptional items (insurance settlement) buried in other income. Encourage students to scan the entire profit and loss account, not just the line highlighted in press commentary.

Case 4: Identifying Sustainable Operating Cash Flow

Learning Objective: Distinguish accounting profit from cash generated by normal business operations.

Case Situation

Bright Home Appliances Ltd, a consumer-durables manufacturer, reported net income of ₹120 crore for the year, up 15% over the previous year — a result the management has highlighted prominently. However, during the same year, trade receivables increased by ₹40 crore and inventory increased by ₹35 crore, while trade payables increased by ₹10 crore. Depreciation for the year was ₹25 crore. A portfolio manager reviewing the results wants to know how much cash the business actually generated from operations before drawing conclusions about earnings quality.

Required

10. Estimate operating cash flow using the indirect method.

11. Compare operating cash flow with reported net income.
12. Comment on the quality of the reported earnings.

Solution

A. Relevant Valuation Principle: Operating cash flow adjusts accounting profit for non-cash items and changes in working capital, revealing how much cash the core business actually generated.

B. Formula: OCF = Net Income + Depreciation − Increase in Receivables − Increase in Inventory + Increase in Payables

C. Substitution and Working:

OCF = 120 + 25 − 40 − 35 + 10 = ₹80 crore

D. Final Result: Operating Cash Flow = ₹80 crore, against reported Net Income of ₹120 crore.

E. Interpretation: OCF is only about 67% of net income. The shortfall is entirely explained by a ₹65 crore net build-up in working capital (receivables and inventory rising faster than payables), meaning a large part of the reported profit is sitting in unsold stock and uncollected customer dues rather than in cash.

F. Decision/Recommendation: The portfolio manager should treat the reported profit growth with some caution and investigate whether the receivable build-up reflects aggressive channel-stuffing (dealer push) or genuine demand growth with a lag in collection, before revising the investment view.

G. Teaching Insight: A common error is ignoring the sign convention — an increase in a current asset is a cash outflow, while an increase in a current liability is a cash inflow. This case is a good opportunity to reinforce that distinction explicitly.

Case 5: Reconstructing Free Cash Flow from Financial Statements

Learning Objective: Convert operating cash flow into free cash flow after considering essential reinvestment.

Case Situation

Suraksha Packaging Ltd, an Indian flexible-packaging manufacturer, reported cash flow from operations of ₹150 crore for the year. During the year, the company incurred capital expenditure of ₹90 crore on plant and machinery, necessary to sustain and modestly expand production capacity, and separately received ₹10 crore from the disposal of an old, idle unit. An analyst preparing a valuation note must decide how to treat each of these items while estimating the company's free cash flow.

Required

13. Estimate the company's free cash flow for the year.
14. Identify which cash flow items are essential reinvestment versus non-recurring/discretionary in nature.
15. Explain why the asset-disposal proceeds should not be treated as part of sustainable free cash flow.

Solution

A. Relevant Valuation Principle: Free cash flow reflects cash generated after reinvestment necessary to sustain operations; one-time, non-operating cash flows should be excluded when estimating a sustainable free cash flow figure for valuation.

B. Formula: Free Cash Flow = Cash Flow from Operations − Capital Expenditure (essential reinvestment)

C. Substitution and Working:

FCF = 150 − 90 = ₹60 crore

(The ₹10 crore disposal proceeds are excluded, as they are a one-time, non-operating inflow.)

D. Final Result: Free Cash Flow = ₹60 crore (before considering the one-time ₹10 crore disposal proceeds).

E. Interpretation: The ₹90 crore capital expenditure is essential reinvestment — required to sustain and modestly grow capacity — and cannot be treated as discretionary. The ₹10 crore disposal proceeds are incidental and non-recurring; including them would overstate the company's ongoing, sustainable free cash flow generation.

F. Decision/Recommendation: The analyst should use ₹60 crore as the sustainable free cash flow figure in the valuation model, and mention the ₹10 crore disposal proceeds separately as a one-time item, not as part of the recurring cash flow base.

G. Teaching Insight: Students often mechanically add every cash-flow-statement line into 'free cash flow.' Reinforce that FCF is a valuation construct built around sustainable, recurring flows — not a literal sum of every reported cash movement.

Case 6: Matching Cash Flow and Discount Rate

Learning Objective: Apply the consistency principle between the type of cash flow and the appropriate discount rate.

Case Situation

Two analysts at a domestic mutual fund are independently valuing Harit Urja Ltd, a renewable-energy company. Analyst A discounts the company's FCFF at the cost of equity of 13%. Analyst B discounts the company's FCFE at the WACC of 10.5%. Both use the same growth rate and arrive at very different values, prompting a debate at the investment desk about which approach, if either, is correct.

Required

16. Identify the inconsistency in each analyst's approach.
17. State the correct pairing of cash flow and discount rate.
18. Explain the likely direction of valuation error in each analyst's estimate.

Solution

A. Relevant Valuation Principle: FCFF, which belongs to all capital providers, must be discounted at the WACC; FCFE, which belongs only to equity holders, must be discounted at the cost of equity. Mismatching cash flow and discount rate produces a systematically biased valuation.

B. Formula: Value = CF_1 / (r − g); the correct pairing is FCFF/WACC and FCFE/Cost of Equity.

C. Substitution and Working:

Analyst A: uses FCFF (belongs to all capital providers) but discounts at cost of equity (13%) instead of WACC (10.5%) — a higher discount rate than appropriate.

Analyst B: uses FCFE (belongs only to equity) but discounts at WACC (10.5%) instead of cost of equity (13%) — a lower discount rate than appropriate.

Since Value = $CF_1/(r - g)$ is inversely related to r, a higher-than-appropriate r understates value, and a lower-than-appropriate r overstates value.

D. Final Result: Analyst A's mismatch (FCFF at cost of equity) understates value; Analyst B's mismatch (FCFE at WACC) overstates value.

E. Interpretation: Both analysts have compounded the same underlying error from opposite directions — each has picked a discount rate belonging to the other cash flow. The correct pairings are FCFF with WACC (value to all capital providers) and FCFE with cost of equity (value to equity holders alone).

F. Decision/Recommendation: The investment desk should reject both estimates and rerun the valuation using FCFF discounted at WACC (or, equivalently, FCFE discounted at cost of equity) before taking an investment view on Harit Urja Ltd.

G. Teaching Insight: This is one of the most common errors in student (and even practitioner) DCF work. A quick memory aid: 'FCFF pays everyone, so it's discounted at the cost of financing everyone (WACC); FCFE pays only equity, so it's discounted at what equity alone requires.'

SECTION II

Dividend Discount Models and Growth Opportunities

Case 7: Stable-Growth Dividend Discount Model

Learning Objective: Estimate the value of a mature dividend-paying company using the constant-growth DDM.

Case Situation

Amrutha Foods Ltd, a long-established NSE-listed FMCG company, paid a dividend of ₹12.00 per share in the year just ended. The company's dividend is expected to grow at a stable 7% per year indefinitely, reflecting its mature, low-volatility earnings profile. The required return on equity for a company of this risk profile is estimated at 12%. The stock currently trades at ₹280 per share, and a mutual fund's equity desk wants an independent estimate of intrinsic value before deciding whether to add to the position.

Required

19. Estimate the intrinsic value per share using the constant-growth dividend discount model.
20. Compare intrinsic value with the current market price.
21. Provide an investment recommendation.

Solution

A. Relevant Valuation Principle: A mature, stable dividend-paying company with a sustainable long-term growth rate below the required return is a natural candidate for the constant-growth (Gordon growth) DDM.

B. Formula: V0 = D1 / (Ke − g), where D1 = D0(1+g)

C. Substitution and Working:

D1 = 12.00 × 1.07 = ₹12.84

V0 = 12.84 / (0.12 − 0.07) = 12.84 / 0.05 = ₹256.80

D. Final Result: Intrinsic value ≈ ₹256.80 per share, against a current market price of ₹280.00.

E. Interpretation: The market price exceeds the DDM-based intrinsic value by about 9.03%, suggesting the stock is priced slightly above what its stable dividend-growth profile alone would justify — possibly reflecting expectations of a re-rating, additional growth avenues, or simply a modest premium the market is willing to pay for earnings stability.

F. Decision/Recommendation: Given the modest overvaluation and the absence of a meaningful margin of safety, the desk should Hold existing holdings but avoid adding fresh exposure at the current price.

G. Teaching Insight: A frequent student error is comparing D1 with market price directly, or using D0 instead of D1 in the numerator. Reinforce that the growth-adjusted next dividend, not the trailing dividend, belongs in the formula.

Case 8: Sustainable Growth in the DDM

Learning Objective: Estimate growth from the retention ratio and ROE rather than accepting an unsupported growth assumption.

Case Situation

Varna Paints Ltd, a listed decorative-paints manufacturer, reported an EPS of ₹20.00 for the year and maintains a dividend payout ratio of 40%. The company's return on equity has been stable at 18%, and its cost of equity is estimated at 14%. Rather than accept a growth figure suggested by a brokerage report, the fund's research head wants the sustainable growth rate derived directly from the company's retention and profitability, and a justified share value based on it.

Required

22. Estimate the sustainable growth rate from the retention ratio and ROE.
23. Compute the justified share value using the constant-growth DDM.
24. Comment on whether the dividend policy supports the growth expectation.

Solution

A. Relevant Valuation Principle: Sustainable growth (g = b × ROE) links a company's reinvestment policy and profitability to the dividend growth it can support without external equity funding.

B. Formula: g = b × ROE, where b = 1 − payout ratio; V0 = D1 / (Ke − g)

C. Substitution and Working:

b = 1 − 0.40 = 0.60

g = 0.60 × 0.18 = 10.80%

D1 = 20.00 × 0.40 × 1.108 = ₹8.864

V0 = 8.864 / (0.14 − 0.108) = 8.864 / 0.032 = ₹277.00

D. Final Result: Sustainable growth = 10.80%; justified share value ≈ ₹277.00.

E. Interpretation: Because ROE (18%) comfortably exceeds the cost of equity (14%), retaining 60% of earnings is genuinely value-accretive; the 10.80% growth rate is therefore internally consistent with, not merely assumed independently of, the company's payout policy.

F. Decision/Recommendation: The 40% payout policy can be treated as sustainable and supportive of the implied growth; the justified value of ₹277.00 can be used as the basis for comparison with the market price.

G. Teaching Insight: Emphasize that g must be derived from b and ROE, not taken as an external analyst's guess disconnected from the company's actual reinvestment economics — this case rewards internal consistency over borrowed growth assumptions.

Case 9: Present Value of Growth Opportunities

Learning Objective: Separate the value of assets in place from the present value of future growth opportunities.

Case Situation

Sanjeevani Pharma Ltd, an Indian generics manufacturer, is expected to post an EPS of ₹30.00 next year and pay a dividend of ₹12.00 per share. Its return on equity is estimated at 20%, and its cost of equity at 15%. An investment committee member wants to understand how much of the company's justified share value comes from its existing earning power versus the value created by expected future reinvestment, before approving a larger position.

Required

1. Estimate the no-growth (assets-in-place) value per share.
2. Estimate the total justified share value under the constant growth DDM.
3. Calculate and interpret the present value of growth opportunities (PVGO).

Solution

A. Relevant Valuation Principle: Total intrinsic value can be decomposed into a no-growth component and PVGO. The no-growth component is next year's EPS capitalized at the required return — this represents the value of the firm *if it paid out 100% of its earnings as dividends and therefore neither grew nor shrank*. PVGO is the additional value created by the firm's *actual* decision to retain and reinvest earnings, over and above simply distributing them.

B. Formula: No-growth value = EPS_1 / Ke *(this assumes a 100% payout — all earnings distributed, zero reinvestment, zero growth)*; Total value = D_1 / (Ke − g); PVGO = Total value − No-growth value

C. Substitution and Working:

Payout = 12/30 = 40%, so retention b = 60%, and g = b × ROE = 0.60 × 0.20 = 12%

No-growth value = 30.00 / 0.15 = ₹200.00 *(value if the firm instead paid out the full ₹30.00 EPS and reinvested nothing)*

Total value = 12.00 / (0.15 − 0.12) = 12.00 / 0.03 = ₹400.00 *(value under the firm's actual policy of retaining 60% and growing at 12%)*

PVGO = 400.00 − 200.00 = ₹200.00

D. Final Result: No-growth value = ₹200.00; total value = ₹400.00; PVGO = ₹200.00.

E. Interpretation: The no-growth value of ₹200.00 is a hypothetical benchmark: it is what the share would be worth if Sanjeevani paid out its entire ₹30.00 EPS each year and never reinvested. The firm instead retains 60% of earnings and reinvests at a 20% ROE — comfortably above its 15% cost of equity — and it is this value-creating reinvestment that lifts the justified value to ₹400.00. The ₹200.00 PVGO therefore represents exactly 50% of the share's total value, meaning half the valuation rests on growth opportunities not yet realized — making it far more sensitive to the durability of the growth pipeline (new launches, capacity, regulatory approvals) than to current earnings alone. Note that PVGO is positive here *only because* the reinvestment rate (ROE 20%) exceeds the cost of equity (15%); had ROE equaled Ke, reinvestment would create no value and PVGO would be zero regardless of the growth rate, and had ROE been below Ke, PVGO would actually be negative.

F. Decision/Recommendation: Before enlarging the position, the committee should specifically stress-test the assumptions behind the 12% growth rate and, critically, the 20% ROE that underpins it, since half of the investment case rests on reinvestment continuing to earn well above the cost of equity — an advantage that competition in generics tends to erode over time.

G. Teaching Insight: Two common errors to pre-empt. First, students often misread the no-growth value E_1/Ke as "the value if the firm keeps its current payout but stops growing" — it is not; it is the value under a *hypothetical 100% payout*. The 60% retention only enters through g and the total-value term. Second, students treat a large PVGO as automatically desirable; in fact it signals that a large portion of value is unproven and vulnerable to disappointment, and it is value-creating *only* when ROE exceeds Ke. PVGO is a measure of future dependence, not a guarantee of quality.

Case 10: Retention Ratio and Shareholder Value

Learning Objective: Understand that higher retention creates value only when the return on reinvested earnings exceeds the required return.

Case Situation

A mid-sized manufacturing company is evaluating three alternative retention policies, all assuming a 60% retention ratio (40% payout) on an EPS of ₹10.00 and a cost of equity of 14%, but under three different ROE scenarios that may apply to the business: 10% (below cost of equity), 14% (equal to cost of equity), and 20% (above cost of equity). The board wants to understand how the same retention policy affects justified value very differently depending on the return earned on reinvested capital.

Required

25. Compute the justified share value and implied P/E under each ROE scenario.
26. Identify in which scenario(s) retaining 60% of earnings creates value.
27. Recommend the payout policy the company should pursue if its true ROE resembles each scenario.

Solution

A. Relevant Valuation Principle: Retaining earnings only creates value when the return earned on the reinvested amount (ROE) exceeds the required return (Ke); when ROE < Ke, retention destroys value regardless of the growth it generates.

B. Formula: g = b × ROE; V0 = D1 / (Ke − g); Implied P/E = V0 / EPS

C. Substitution and Working:

Scenario A (ROE 10%): g = 6.0%, D1 = ₹4.24, V0 = 4.24/0.08 = ₹53.00, P/E = 5.30

Scenario B (ROE 14%): g = 8.4%, D1 = ₹4.336, V0 = 4.336/0.056 = ₹77.43, P/E = 7.74

Scenario C (ROE 20%): g = 12.0%, D1 = ₹4.48, V0 = 4.48/0.02 = ₹224.00, P/E = 22.40

D. Final Result: Justified value/P/E: Scenario A ₹53.00 (5.30x); Scenario B ₹77.43 (7.74x); Scenario C ₹224.00 (22.40x) — all under the identical 60% retention policy.

E. Interpretation: The same 60% retention ratio produces wildly different outcomes purely because of the ROE assumption: value is modest when ROE is below Ke, moderate when ROE equals Ke, and dramatically higher when ROE exceeds Ke. Retention is not inherently good or bad — its effect depends entirely on what the retained capital can earn.

F. Decision/Recommendation: The company should retain a high proportion of earnings only if its ROE genuinely exceeds its cost of equity (Scenario C); if ROE is at or below the cost of equity (Scenarios A and B), a higher payout (lower retention) would serve shareholders better.

G. Teaching Insight: This case is a strong classroom anchor for the retention–ROE–value relationship. A common misconception is that retention always compounds value through growth; this case demonstrates growth can be value-destroying when unsupported by adequate returns.

Case 11: Effect of a Change in Dividend Payout

Learning Objective: Evaluate the combined effect of payout and growth on intrinsic value.

Case Situation

Nirmala Consumer Products Ltd, a cash-rich Indian FMCG company with an EPS of ₹15.00, a return on equity of 16%, and a cost of equity of 13%, is considering raising its dividend payout ratio from 40% to 60% to please income-seeking shareholders. The CFO believes this will make the stock more attractive. The board has asked for a comparison of intrinsic value under both payout policies before the proposal is finalised.

Required

28. Estimate the sustainable growth rate and justified share value at a 40% payout.
29. Estimate the sustainable growth rate and justified share value at a 60% payout.
30. Explain why a larger immediate dividend may not increase intrinsic value.

Solution

A. Relevant Valuation Principle: When ROE exceeds the cost of equity, a higher payout ratio reduces reinvestment, lowers sustainable growth, and can reduce — not raise — intrinsic value, despite handing shareholders a larger current dividend.

B. Formula: g = b × ROE; V0 = D1 / (Ke − g)

C. Substitution and Working:

At 40% payout: b = 60%, g = 9.6%, D1 = ₹6.576, V0 = 6.576/0.034 = ₹193.41

At 60% payout: b = 40%, g = 6.4%, D1 = ₹9.576, V0 = 9.576/0.066 = ₹145.09

D. Final Result: Justified value falls from ₹193.41 (at 40% payout) to ₹145.09 (at 60% payout).

E. Interpretation: Because ROE (16%) exceeds Ke (13%), the earnings retained under the lower-payout policy are being reinvested at a rate that creates more value than the equivalent amount would if distributed as dividends today. Raising the payout therefore trades away future value-creating reinvestment for a larger current cash dividend.

F. Decision/Recommendation: The board should not proceed with the payout increase from 40% to 60% purely to raise shareholder value; if the intent is specifically to satisfy income-seeking investors, that trade-off should be recognised explicitly as reducing intrinsic value.

G. Teaching Insight: Students often assume a bigger dividend is unambiguously good for shareholders; this case demonstrates that payout policy is a value trade-off, not a free lunch, whenever ROE exceeds Ke.

Case 12: Two-Stage Dividend Discount Model

Learning Objective: Value a company experiencing temporary high growth followed by stable growth.

Case Situation

Nidan Diagnostics Ltd, a mid-sized diagnostic-services company, has just paid a dividend of ₹5.00 per share. Given its rapid network expansion, dividends are expected to grow at 20% per year for the next four years before settling into a stable long-term growth rate of 6%. The required return on equity is 15%. An analyst preparing an initiation report needs the present value of both the high-growth dividends and the terminal value beyond year four.

Required

31. Estimate the dividends for years 1 through 4 and the terminal value at the end of year 4.
32. Calculate the present value of the interim dividends and the terminal value.
33. Estimate the intrinsic value per share.

Solution

A. Relevant Valuation Principle: A two-stage DDM values the explicit high-growth dividends separately from a terminal value computed once dividends settle into stable, perpetual growth.

B. Formula: Dt = D0(1+g_high)^t for t = 1..4; TV4 = D5/(Ke − g_stable); V0 = Σ PV(Dt) + PV(TV4)

C. Substitution and Working:
D1=₹6.00, D2=₹7.20, D3=₹8.64, D4=₹10.368

D5 = 10.368 × 1.06 = ₹10.99; TV4 = 10.99/(0.15−0.06) = ₹122.11

PV of dividends (years 1–4, discounted at 15%) = ₹22.27

PV of TV4 = 122.11 × $(1.15)^{-4}$ = ₹69.82

D. Final Result: Intrinsic value = 22.27 + 69.82 ≈ ₹92.09 per share.

E. Interpretation: About 76% of the intrinsic value (₹69.82 of ₹92.09) comes from the terminal value beyond year 4, underscoring how sensitive two-stage DDM valuations are to the stable-growth assumption and the length of the explicit high-growth period, rather than to the near-term dividends alone.

F. Decision/Recommendation: The analyst should stress-test the 6% stable growth and the four-year high-growth window before finalising the ₹92.09 target, given how much of the value rests on those two assumptions.

G. Teaching Insight: A common calculation error is discounting the terminal value back only three years instead of four (from the end of the explicit period, not from year 5). Reinforce that TV is computed at the end of the LAST explicit-period year.

Case 13: High-Growth Phase Followed by a Transition Period

Learning Objective: Incorporate gradually declining growth rather than assuming an abrupt move from high growth to stable growth.

Case Situation

Vidyut EV Components Ltd, an Indian electric-vehicle component maker, has just paid a dividend of ₹4.00 per share. Dividend growth is expected at 25% for the next three years as the company scales up, then decline gradually over three further years to 20%, 15%, and 10%, before settling into a stable long-term growth rate of 6%. The required return on equity is 14%. A portfolio manager wants a value that reflects this more gradual moderation rather than an abrupt drop to stable growth after year three.

Required

34. Project the dividends for years 1 through 6 and the terminal value at the end of year 6.
35. Calculate the present value of the projected dividends and the terminal value.
36. Explain why a gradual transition period is more realistic than an abrupt shift to stable growth.

Solution

A. Relevant Valuation Principle: A transition-period model avoids the unrealistic assumption that a rapidly growing company's growth collapses instantly to a mature-company rate; growth typically moderates gradually as competition, market saturation, and scale effects take hold.

B. Formula: $D_t = D_{(t-1)} \times (1+g_t)$; $TV_6 = D_7/(K_e - g_{stable})$; $V_0 = \Sigma\, PV(D_t) + PV(TV_6)$

C. Substitution and Working:

Dividends (₹): D1=5.00, D2=6.25, D3=7.8125, D4=9.375, D5=10.78, D6=11.86

D7 = 11.86 × 1.06 = ₹12.57; TV6 = 12.57/0.08 = ₹157.14

PV of dividends (years 1–6, at 14%) = ₹31.02

PV of TV6 = 157.14 × $(1.14)^{-6}$ = ₹71.59

D. Final Result: Intrinsic value = 31.02 + 71.59 ≈ ₹102.61 per share.

E. Interpretation: The gradually declining growth path (25% → 20% → 15% → 10% → 6%) avoids overstating near-term dividends relative to a realistic maturation curve, while also avoiding understating them relative to an abrupt three-year cutoff. The resulting value of ₹102.61 sits between what a pure high-growth-then-abrupt-stable model and an overly conservative model would produce.

F. Decision/Recommendation: The portfolio manager can treat ₹102.61 as a more defensible base-case value than either a naive constant-25%-then-6% model or an overly compressed growth assumption, and should use it as the reference point for comparison with the market price.

G. Teaching Insight: Emphasise to students that transition periods exist precisely because real businesses rarely experience a sudden step-change in growth; modelling a gradual glide path is a meaningfully different (and usually more defensible) assumption than a hard two-stage cutoff.

Case 14: Three-Stage Dividend Discount Model

Learning Objective: Apply different growth rates across high-growth, transition, and stable stages.

Case Situation

A branded apparel company has just paid a dividend of ₹6.00 per share. Dividends are projected to grow at 20% for the next four years, then 13% for the following four years, before settling into a stable long-term growth rate of 7%. The required return on equity is 15%. An analyst preparing a full valuation report wants to identify not only the intrinsic value but which stage of growth contributes most to that value.

Required

37. Project dividends for the eight explicit years and estimate the terminal value at the end of year 8.
38. Calculate the present value contribution of each of the three stages (high-growth, transition, and terminal value).
39. Estimate the intrinsic value and identify the stage contributing most to value.

Solution

A. Relevant Valuation Principle: A three-stage DDM separately values a high-growth phase, a transition phase at a different (but still above-stable) growth rate, and a terminal value once growth stabilises.

B. Formula: Dt computed sequentially by stage; TV8 = D9/(Ke − g_stable); V0 = PV(stage 1) + PV(stage 2) + PV(TV8)

C. Substitution and Working:

Years 1–4 (20% growth): dividends ₹7.20, 8.64, 10.37, 12.44; PV (stage 1) ≈ ₹26.72

Years 5–8 (13% growth): dividends ₹14.06, 15.89, 17.95, 20.29; PV (stage 2) ≈ ₹27.24

D9 = 20.29 × 1.07 = ₹21.71; TV8 = 21.71/0.08 = ₹271.32; PV(TV8) ≈ ₹88.70

D. Final Result: Intrinsic value = 26.72 + 27.24 + 88.70 ≈ ₹142.66 per share.

E. Interpretation: The terminal value alone contributes about 62% of total value (₹88.70 of ₹142.66), with the high-growth and transition stages contributing roughly 19% and 19% respectively — a reminder that even with an explicit eight-year forecast, most of the value still depends on the stable-growth assumption beyond the modelled horizon.

F. Decision/Recommendation: Given the terminal value's dominant share of total value, the analyst should present the ₹142.66 estimate together with a clear sensitivity range around the 7% stable-growth assumption rather than as a single-point figure.

G. Teaching Insight: Use this case to show that adding more explicit-forecast years does not eliminate dependence on the terminal-value assumption — it merely defers it. Students often mistakenly believe a longer forecast period reduces terminal-value sensitivity; in practice the terminal value typically still dominates.

Case 15: EPS, Payout, and Multistage Dividends

Learning Objective: Derive dividends from forecast EPS and payout ratios during different growth stages.

Case Situation

A specialty-chemicals company currently has an EPS of ₹10.00, expected to grow at 18% annually over the next three years, while it pays out only 20% of earnings during this expansion phase. Thereafter, EPS growth is expected to stabilize at 7%, supported by a sustainable ROE of 16%, with the payout ratio rising to whatever level is consistent with that stable growth. The required return on equity is 14%. Students are given only the EPS growth, payout, and ROE assumptions — not the dividend figures directly.

Required

40. Derive the EPS and dividend for each of the three high-growth years.
41. Derive the stable-stage payout ratio consistent with the given ROE and stable growth, and the terminal value.
42. Estimate the intrinsic value per share.

Solution

A. Relevant Valuation Principle: Dividends must be derived from EPS and payout assumptions at each stage; the stable-stage payout should itself be internally consistent with the stable growth rate and stable ROE (via g = b × ROE), not assumed independently.

B. Formula: Dt = EPSt × payout_t; stable b = g_stable/ROE_stable; TV = D(stable) / (Ke − g_stable)

C. Substitution and Working:

EPS: Year1 ₹11.80, Year2 ₹13.92, Year3 ₹16.43; Dividends (20% payout): ₹2.36, ₹2.78, ₹3.29

Stable-stage retention b = 7%/16% = 43.75%, so stable payout = 56.25%

EPS4 = 16.43 × 1.07 = ₹17.58; D4 = 17.58 × 0.5625 = ₹9.89; TV3 = 9.89/(0.14−0.07) = ₹141.27

PV of dividends (years 1–3, at 14%) ≈ ₹6.43; PV of TV3 ≈ ₹95.35

D. Final Result: Intrinsic value ≈ 6.43 + 95.35 ≈ ₹101.79 per share.

E. Interpretation: The low 20% payout during the high-growth phase means the near-term dividends contribute very little to value; almost 94% of the intrinsic value depends on the terminal value, which in turn depends on the stable-stage payout being derived consistently (56.25%) from the given ROE and growth, rather than assumed arbitrarily.

F. Decision/Recommendation: The valuation of ₹101.79 should be presented with an explicit note on how sensitive it is to the stable-stage ROE and growth assumptions, since the explicit high-growth dividends are a minor part of total value.

G. Teaching Insight: This case tests whether students can derive the internally consistent stable payout ratio (via b = g/ROE) rather than picking an arbitrary number; a common error is assuming a payout ratio disconnected from the growth and ROE already specified.

Case 16: DDM Suitability for an Irregular Dividend Company

Learning Objective: Judge when the dividend discount model is conceptually inappropriate even though dividends are observable.

Case Situation

A promoter-controlled engineering company has paid dividends of ₹2.00, ₹6.00, ₹1.50, and ₹3.00 per share over the last four years — a pattern driven largely by the promoter group's personal liquidity needs rather than the company's earnings capacity or investment opportunities. The most recent dividend was ₹3.00 per share, and the cost of equity is estimated at 13%. An analyst has been asked to provide a quick DDM-based estimate but is uneasy about its reliability.

Required

43. Compute an indicative DDM value using the latest dividend and a zero-growth assumption.
44. Identify why this DDM estimate is likely to be misleading.
45. Recommend a more appropriate valuation approach.

Solution

A. Relevant Valuation Principle: The DDM assumes dividends are a reasonable proxy for distributable earnings power; when dividends are erratic and disconnected from earnings and investment policy, the model's core assumption breaks down.

B. Formula: V0 (naive, zero-growth) = D0 / Ke

C. Substitution and Working:

V0 = 3.00 / 0.13 = ₹23.08

D. Final Result: A naive DDM estimate gives ≈ ₹23.08 per share, but this figure should be treated with considerable skepticism.

E. Interpretation: Because the dividend history (₹2.00 → ₹6.00 → ₹1.50 → ₹3.00) shows no relationship to a stable payout or growth pattern, any DDM value built on the latest or an averaged dividend is essentially arbitrary — it reflects the promoter group's cash needs, not the company's sustainable earning power or reinvestment economics.

F. Decision/Recommendation: The analyst should not rely on the DDM for this company; a free-cash-flow-to-equity (FCFE) approach, which values cash available for distribution rather than cash actually distributed, would better reflect the company's underlying earning power regardless of the promoter's payout choices.

G. Teaching Insight: This case is a useful check on students' tendency to apply the DDM mechanically whenever a dividend figure exists. Reinforce that model choice should be driven by whether the model's assumptions plausibly hold, not merely by data availability.

Case 17: Market Price, DDM Value, and Recommendation Risk

Learning Objective: Combine a DDM estimate with uncertainty surrounding growth and required return.

Case Situation

An equity analyst covering a listed hospital chain has estimated its intrinsic value using the constant growth DDM. The company's most recent dividend was ₹9.00 per share, expected to grow at 8% annually, with a required return on equity of 13%. The stock currently trades at ₹180 per share. Before finalising a recommendation, the analyst wants to identify which of the two key assumptions — growth or required return — the conclusion is most sensitive to.

Required

46. Estimate the intrinsic value per share and the valuation gap versus the market price.
47. Provide an investment recommendation.
48. Identify whether growth or the required return is the more critical assumption, and why.

Solution

A. Relevant Valuation Principle: When the spread between Ke and g is narrow, small changes in either input produce disproportionately large changes in value — the valuation conclusion rests more on the reliability of the spread than on either input in isolation.

B. Formula: V0 = D1/(Ke − g); gap = (V0 − CMP)/CMP

C. Substitution and Working:

D1 = 9.00 × 1.08 = ₹9.72

V0 = 9.72/(0.13 − 0.08) = 9.72/0.05 = ₹194.40

Gap = (194.40 − 180)/180 = 8.00%

D. Final Result: Intrinsic value ≈ ₹194.40, a moderate 8.00% premium to the ₹180 market price.

E. Interpretation: Because the Ke − g spread is only 5 percentage points, a shift of even 1 percentage point in either growth or required return would change the spread by 20% and move the value by a similar order of magnitude — the ₹194.40 figure is therefore considerably less firm than its precise appearance suggests.

F. Decision/Recommendation: A cautious Buy can be supported given the positive but moderate margin of safety, but the recommendation should flag that the conclusion is highly sensitive to the narrow Ke − g spread rather than presenting ₹194.40 as a precise target.

G. Teaching Insight: Use this case to illustrate that a narrow discount-rate/growth spread is itself a risk factor worth disclosing, independent of the direction of the valuation gap. Sensitivity commentary should accompany any DDM value built on a small spread.

SECTION III

FCFF, FCFE, and Explicit-Period Cash Flows

Case 18: Estimating FCFF from Operating Profit

Learning Objective: Calculate unlevered free cash flow from EBIT, tax, depreciation, capital expenditure, and working-capital investment.

Case Situation

A listed cement manufacturer is undertaking a capacity-expansion project. For the forecast year, the company expects EBIT of ₹200 crore, a corporate tax rate of 25%, depreciation of ₹40 crore, capital expenditure of ₹90 crore, and an increase in working capital of ₹15 crore. The corporate-finance team wants the firm's free cash flow to the firm (FCFF) for the year, independent of how the expansion is financed.

Required

49. Calculate FCFF for the forecast year.
50. Explain why interest expense is excluded from the FCFF calculation.

Solution

A. Relevant Valuation Principle: FCFF represents cash flow available to all capital providers (both debt and equity) before financing effects, so it must be computed on an unlevered (pre-interest) basis.

B. Formula: FCFF = EBIT(1 − t) + Depreciation − Capital Expenditure − Increase in Working Capital

C. Substitution and Working:
FCFF = 200 × 0.75 + 40 − 90 − 15 = 150 + 40 − 90 − 15 = ₹85 crore

D. Final Result: FCFF = ₹85 crore for the forecast year.

E. Interpretation: FCFF of ₹85 crore represents cash generated by operations and available to both lenders and shareholders, before any consideration of how the ₹90 crore capex is actually funded (debt, equity, or internal accruals).

F. Decision/Recommendation: This ₹85 crore figure, not a levered cash-flow number, should be used as the basis for a WACC-discounted enterprise valuation of the expansion.

G. Teaching Insight: Interest expense is excluded because FCFF measures the cash generated by the business regardless of capital structure; including interest would mix an operating measure with a financing decision, and would double-count the cost of debt (once in the cash flow, once in the discount rate).

Case 19: Estimating FCFE from Net Income

Learning Objective: Estimate cash flow available to equity shareholders after reinvestment and net borrowing.

Case Situation

An Indian consumer-finance company reported net income of ₹60 crore for the year. Depreciation for the year was ₹8 crore, capital expenditure was ₹20 crore, and working capital increased by ₹5 crore. The company also raised fresh borrowings of ₹25 crore and repaid ₹10 crore of existing debt during the year. The board wants to know whether the company generated enough free cash flow to equity to fund its planned dividend without needing additional external borrowing.

Required

51. Calculate free cash flow to equity (FCFE) for the year.
52. Assess whether the company can fund its dividends from FCFE without additional borrowing.

Solution

A. Relevant Valuation Principle: FCFE measures cash available specifically to equity shareholders after meeting reinvestment needs and after accounting for net cash flows to and from debt holders.

B. Formula: FCFE = Net Income + Depreciation − Capital Expenditure − Increase in Working Capital + Net Borrowing (New Debt − Debt Repaid)

C. Substitution and Working:

Net borrowing = 25 − 10 = ₹15 crore

FCFE = 60 + 8 − 20 − 5 + 15 = ₹58 crore

D. Final Result: FCFE = ₹58 crore for the year.

E. Interpretation: With FCFE of ₹58 crore, the company has generated more than enough cash, after reinvestment and net debt movements, to support a meaningful dividend without needing to raise additional external capital, as long as the actual dividend declared stays within this amount.

F. Decision/Recommendation: The board can approve a dividend up to ₹58 crore in aggregate (subject to regulatory and prudential capital requirements applicable to a consumer-finance company) without requiring fresh borrowing.

G. Teaching Insight: A common student error is omitting net borrowing from the FCFE calculation, effectively treating FCFE as identical to FCFF adjusted only for interest. Stress that net new borrowing is a genuine source of cash available to equity holders in the FCFE framework.

Case 20: Reconciliation between FCFF and FCFE

Learning Objective: Understand the mathematical and economic connection between firm cash flow and equity cash flow.

Case Situation

A manufacturing company has an estimated FCFF of ₹100 crore for the year. It pays interest expense of ₹20 crore, faces a corporate tax rate of 25%, and raised net new borrowing of ₹10 crore during the year. An analyst who has already computed FCFF now needs to derive FCFE for the same company without recomputing cash flow from scratch.

Required

53. Derive FCFE from the given FCFF using the appropriate reconciliation formula.
54. Explain how debt financing changes the cash flow attributable to shareholders without changing the underlying operating cash generation.

Solution

A. Relevant Valuation Principle: FCFE can be derived directly from FCFF by removing the after-tax cost of debt service and adding back new net borrowing, since both cash flows share the same underlying operating cash generation.

B. Formula: FCFE = FCFF − Interest (1 − t) + Net Borrowing

C. Substitution and Working:

FCFE = 100 − 20 × 0.75 + 10 = 100 − 15 + 10 = ₹95 crore

D. Final Result: FCFE = ₹95 crore, derived from the given FCFF of ₹100 crore.

E. Interpretation: The operating business generates ₹100 crore in FCFF regardless of financing choices; how much of that ultimately belongs to equity holders (₹95 crore here) depends on the after-tax interest cost paid to lenders and the extent to which fresh borrowing supplements the residual.

F. Decision/Recommendation: The analyst can use the ₹95 crore FCFE figure directly in an equity-value (FCFE-based) valuation without needing to rebuild the cash-flow forecast from operating statements.

G. Teaching Insight: This reconciliation is a useful check when FCFF has already been estimated elsewhere in a model; a common error is forgetting to tax-affect the interest expense before deducting it or omitting net borrowing entirely.

Case 21: Levered versus Unlevered Free Cash Flow

Learning Objective: Distinguish cash flows before financing from cash flows after financing.

Case Situation

Two analysts are examining an infrastructure developer that raised significant fresh debt during the year to fund a highway project. The company's EBIT is ₹150 crore, the tax rate is 25%, depreciation is ₹30 crore, capital expenditure is ₹80 crore, and working capital increased by ₹10 crore. Interest expense for the year was ₹25 crore, and the company raised a new net debt of ₹40 crore. One analyst reports a much higher 'free cash flow' figure than the other, attributing the difference to the fresh borrowing, and claims this reflects stronger value creation.

Required

55. Classify each given item as operating or financing in nature.
56. Estimate both FCFF and FCFE for the year.
57. Explain why the additional debt raised does not represent operating value creation.

Solution

A. Relevant Valuation Principle: FCFF captures only operating cash generation, independent of financing; FCFE additionally reflects financing cash flows (interest and net borrowing), which can inflate the equity cash flow without any change in the underlying business's operating performance.

B. Formula: FCFF = EBIT(1−t) + Depreciation − Capex − ΔWC; FCFE = FCFF − Interest(1−t) + Net Borrowing

C. Substitution and Working:

FCFF = 150 × 0.75 + 30 − 80 − 10 = 112.5 + 30 − 80 − 10 = ₹52.50 crore

FCFE = 52.50 − 25 × 0.75 + 40 = 52.50 − 18.75 + 40 = ₹73.75 crore

D. Final Result: FCFF = ₹52.50 crore (unlevered/operating); FCFE = ₹73.75 crore (levered, after financing).

E. Interpretation: FCFE (₹73.75 crore) exceeds FCFF (₹52.50 crore) purely because of the ₹40 crore net new borrowing — the underlying operating business generated only ₹52.50 crore. The analyst reporting the higher levered figure has mistaken a financing inflow for stronger operating performance.

F. Decision/Recommendation: Any assessment of the project's operating value creation should be based on the ₹52.50 crore FCFF, discounted at WACC; the ₹73.75 crore FCFE reflects financing choices and should not be presented as evidence of superior business performance.

G. Teaching Insight: This case directly confronts a common analytical error: treating a debt-funded increase in equity cash flow as value creation. Stress that raising debt does not create operating value — it merely reallocates and can temporarily inflate near-term cash available to equity.

Case 22: Forecasting Intermediate-Period Revenue and Margin

Learning Objective: Translate operating assumptions into explicit-period free cash flows.

Case Situation

An Indian organized retail chain, currently generating revenue of ₹500 crore, is opening new stores over the next three years. Revenue is expected to grow 20%, 22%, and 18% in years 1, 2, and 3, respectively, with the operating margin (EBIT margin) improving from 8% in year 1 to 9% in year 2 and 10% in year 3 as new stores mature. The tax rate is 25%. Depreciation is expected at ₹15, ₹18, and ₹20 crore; capital expenditure at ₹25, ₹28, and ₹22 crore; and working-capital investment at ₹10, ₹12, and ₹8 crore for years 1, 2, and 3 respectively. The CFO wants a compact three-year FCFF forecast.

Required

58. Forecast revenue for each of the three years.
59. Estimate FCFF for each of the three years using the given margin, tax, depreciation, capex, and working-capital assumptions.

Solution

A. Relevant Valuation Principle: Explicit-period free cash flow forecasts are built up year by year from revenue and margin assumptions, converted through tax, and adjusted for reinvestment needs — not assumed as a flat percentage of revenue.

B. Formula: $Revenue_t = Revenue_{(t-1)} \times (1+growth_t)$; $FCFF_t = Revenue_t \times Margin_t \times (1-t) + Dep_t - Capex_t - \Delta WC_t$

C. Substitution and Working:

Revenue: Year1 ₹600.00 cr, Year2 ₹732.00 cr, Year3 ₹863.76 cr

FCFF Year1 = $600 \times 0.08 \times 0.75 + 15 - 25 - 10 = 36.00 + 15 - 25 - 10 =$ ₹16.00 cr

FCFF Year2 = $732 \times 0.09 \times 0.75 + 18 - 28 - 12 = 49.41 + 18 - 28 - 12 =$ ₹27.41 cr

FCFF Year3 = $863.76 \times 0.10 \times 0.75 + 20 - 22 - 8 = 64.78 + 20 - 22 - 8 =$ ₹54.78 cr

D. Final Result: FCFF forecast: Year1 ₹16.00 crore, Year2 ₹27.41 crore, Year3 ₹54.78 crore.

E. Interpretation: FCFF grows much faster than revenue over the three years (from ₹16.00 crore to ₹54.78 crore, a more than threefold increase) because margin improvement and moderating capital intensity compound on top of revenue growth as new stores mature and required reinvestment eases relative to the expanding revenue base.

F. Decision/Recommendation: This three-year FCFF path, together with an appropriate terminal value from year 4 onward, can now be discounted at the retailer's WACC to estimate enterprise value.

G. Teaching Insight: This case reinforces that explicit-period forecasting requires linking each cash-flow driver (margin, capex, working capital) to the underlying business narrative (new-store maturation) rather than applying a single blended assumption across all years.

Case 23: Growth, Reinvestment, and Free Cash Flow

Learning Objective: Recognize that rapid earnings growth may initially reduce free cash flow because of heavy reinvestment.

Case Situation

A solar-module manufacturer, currently generating revenue of ₹300 crore, expects strong sales growth of 35% and 30% over the next two years as demand accelerates. Operating margin is expected to improve from 10% to 12% over this period, aided by scale, with a 25% tax rate. However, the expansion requires substantial capital expenditure of ₹45 crore and ₹50 crore, along with inventory and receivables build-up (working-capital investment) of ₹20 crore and ₹18 crore, while depreciation is ₹10 crore and ₹14 crore in years 1 and 2, respectively. Management is puzzled by FCFF's weakness despite strong reported profit growth.

Required

60. Estimate NOPAT (after-tax operating profit) for each of the two years.
61. Estimate FCFF for each of the two years.
62. Explain why accounting profit growth and cash-flow growth are moving in different directions.

Solution

A. Relevant Valuation Principle: Rapid revenue growth often requires disproportionately heavy reinvestment in capacity and working capital, which can cause free cash flow to be low or even negative, even as accounting profit (NOPAT) rises strongly.

B. Formula: $NOPAT_t = Revenue_t \times Margin_t \times (1-t)$; $FCFF_t = NOPAT_t + Dep_t - Capex_t - \Delta WC_t$

C. Substitution and Working:

Revenue: Year1 ₹405.00 cr, Year2 ₹526.50 cr

NOPAT: Year1 = 405×0.10×0.75 = ₹30.38 cr; Year2 = 526.5×0.12×0.75 = ₹47.39 cr

FCFF Year1 = 30.38+10−45−20 = −₹24.63 cr

FCFF Year2 = 47.39+14−50−18 = −₹6.62 cr

D. Final Result: NOPAT rises from ₹30.38 crore to ₹47.39 crore (+56%), while FCFF remains negative in both years: −₹24.63 crore and −₹6.62 crore.

E. Interpretation: Despite strong and accelerating accounting profitability, the company is consuming more cash in capex and working capital than it generates from operations in both years — a classic pattern for a fast-scaling manufacturer, where growth is genuinely value-creating for the long run but cash-flow-negative in the near term.

F. Decision/Recommendation: Management should not treat the negative FCFF as a warning sign in isolation; instead, the investment case should be evaluated on whether the reinvestment is generating adequate future returns, with FCFF expected to turn positive as growth moderates and the capex program tapers.

G. Teaching Insight: This case corrects a common misconception that negative free cash flow always signals distress. High-growth, high-reinvestment businesses routinely show negative FCFF alongside strong, rising accounting profits — the two measures answer different questions.

Case 24: Positive FCFF but Negative FCFE

Learning Objective: Explain how debt repayment can produce negative equity cash flow despite positive operating cash flow.

Case Situation

A telecom-tower company generates a stable FCFF of ₹180 crore for the year, reflecting healthy, predictable operating cash generation from long-term tenancy contracts. However, the company is committed to a large scheduled net debt repayment of ₹100 crore during the year as part of its deleveraging plan, in addition to interest expense of ₹40 crore at a 25% tax rate. The finance team is concerned about what this means for cash available to equity shareholders.

Required

63. Calculate FCFE for the year.
64. Assess whether the resulting FCFE necessarily indicates operating weakness.

Solution

A. Relevant Valuation Principle: FCFE can turn negative even when FCFF is healthy and positive, if scheduled net debt repayment (a financing outflow) exceeds the equity cash flow otherwise available after interest.

B. Formula: FCFE = FCFF − Interest (1 − t) + Net Borrowing (negative when net debt is repaid)

C. Substitution and Working:

FCFE = 180 − 40 × 0.75 + (−100) = 180 − 30 − 100 = ₹50.00 crore

D. Final Result: FCFE = ₹50.00 crore for the year (positive, though well below the ₹180 crore FCFF).

E. Interpretation: Although FCFE remains positive here, it is dramatically lower than FCFF purely because of the scheduled debt repayment — illustrating how a large net debt repayment can push FCFE sharply lower, or even negative in years with larger repayments, without any deterioration in the underlying operating business.

F. Decision/Recommendation: The finance team should not interpret a low or even negative FCFE in a deleveraging year as a sign of operating weakness; the appropriate check is whether FCFF remains healthy and stable, which it does here at ₹180 crore.

G. Teaching Insight: This case reinforces that FCFE reflects financing choices (including voluntary or scheduled deleveraging) as much as operating performance. A common error is treating a weak or negative FCFE figure alone as an operating red flag without checking the underlying FCFF.

Case 25: Choosing FCFF under Changing Leverage

Learning Objective: Select FCFF when the debt-equity mix is expected to change materially.

Case Situation

A road infrastructure company plans to steadily reduce its debt levels over the next five years, using toll revenues to pay down project debt ahead of schedule. An analyst is deciding whether to value the company using FCFE (discounted at the cost of equity) or FCFF (discounted at WACC), given that both the capital structure and, consequently, the cost of equity will change materially over the forecast period.

Required

65. Recommend whether FCFF or FCFE is more suitable for this valuation and explain why.
66. Identify why estimating a changing cost of equity and net-borrowing path each year would complicate an FCFE-based valuation.

Solution

A. Relevant Valuation Principle: FCFF valued at WACC is generally more robust when the debt-equity mix is expected to change materially over the forecast period, since it avoids the need to re-estimate the shifting cost of equity and net-borrowing schedule at every stage.

B. Formula: (Conceptual case — no single numerical formula; comparison of FCFF/WACC versus FCFE/Ke suitability.)

C. Substitution and Working:

Under FCFE, the cost of equity would need to be re-levered each year as leverage falls (declining beta, declining Ke), and net borrowing (repayment) would need forecasting precisely each year — both introduce compounding estimation risk. Under FCFF, WACC can be estimated using a representative capital structure (e.g., average or target), and the debt-repayment schedule is captured indirectly through the debt figure used in the final equity-value bridge, rather than through the cash-flow forecast itself.

D. Final Result: FCFF, discounted at WACC, is the more suitable approach for this company.

E. Interpretation: With debt falling steadily over five years, FCFE would require a fresh cost-of-equity estimate (via relevering beta) for each forecast year and a precise net-borrowing forecast — both significant sources of estimation error. FCFF sidesteps this by valuing operations independent of the financing path, with the debt-equity bridge applied just once, at the valuation date.

F. Decision/Recommendation: The analyst should adopt FCFF discounted at WACC (using either a single representative WACC or a lightly varying WACC across the changing capital structure), and reserve FCFE for cases where leverage is expected to remain broadly stable.

G. Teaching Insight: This case builds judgment rather than computation: emphasize to students that model choice should be driven by which approach requires fewer unstable, compounding assumptions — not simply by which cash flow is more familiar to compute.

Case 26: FCFE Valuation of a Bank or NBFC

Learning Objective: Understand why equity cash flow is often more meaningful than enterprise cash flow for financial institutions.

Case Situation

An RBI-regulated NBFC expects earnings of ₹80 crore for the year. Regulatory capital adequacy norms require the NBFC to retain earnings equivalent to ₹35 crore to support growth in its loan book, leaving the remainder available for distribution as dividends. The finance team notes that, unlike a manufacturing company, debt for this NBFC is itself a raw material (funds borrowed to be lent onward) rather than a distinct financing choice separate from operations.

Required

67. Estimate the FCFE (distributable cash flow) for the NBFC.
68. Explain why classifying debt as a separate financing item, as is done for FCFF, is difficult for a financial institution.

Solution

A. Relevant Valuation Principle: For banks and NBFCs, debt (borrowed funds) is integral to the core operating activity of lending, making a clean separation between 'operating' and 'financing' cash flows — the basis of FCFF — largely impractical; FCFE, which nets out regulatory capital retention, is the more natural measure.

B. Formula: FCFE (distributable) = Earnings − Regulatory Capital Retention Required for Growth

C. Substitution and Working:

FCFE = 80 − 35 = ₹45 crore

D. Final Result: FCFE (distributable cash flow) = ₹45 crore for the year.

E. Interpretation: Of the ₹80 crore earned, ₹35 crore must be retained purely to maintain the regulatory capital needed to support the growing loan book — this is not discretionary reinvestment in the usual sense but a regulatory necessity, leaving ₹45 crore genuinely available for equity holders.

F. Decision/Recommendation: The ₹45 crore FCFE figure, not an FCFF-based enterprise valuation, should be used as the basis for valuing the NBFC's equity, since separating borrowed funds into a distinct 'financing' layer would misrepresent how the business actually operates.

G. Teaching Insight: A common error is attempting to force a standard FCFF/enterprise-value framework onto a financial institution. Reinforce that for banks and NBFCs, borrowed funds are the raw material of the business (analogous to inventory for a manufacturer), making FCFE the conceptually cleaner starting point.

Case 27: FCFF Valuation and the Equity-Value Bridge

Learning Objective: Convert the present value of operating free cash flows into enterprise value and then equity value.

Case Situation

A listed engineering company is expected to generate FCFF of ₹40 crore, ₹46 crore, ₹53 crore, and ₹61 crore over the next four years, after which cash flow is expected to grow at a stable 5% per year in perpetuity. The company's WACC is 11.5%. It carries a debt of ₹150 crore and cash of ₹40 crore and has 20 crore shares outstanding. A portfolio manager wants the enterprise value, equity value, and value per share, along with a recommendation.

Required

69. Estimate the terminal value at the end of year 4 and the enterprise value.
70. Estimate the equity value and value per share.
71. Provide a buy or avoid recommendation, assuming the current market price is ₹28 per share.

Solution

A. Relevant Valuation Principle: Enterprise value is the present value of all forecast and terminal FCFF; equity value is then derived by subtracting debt and adding cash, before dividing by outstanding shares.

B. Formula: TV4 = FCFF5/(WACC − g); EV = Σ PV(FCFFt) + PV(TV4); Equity Value = EV − Debt + Cash; Value/share = Equity Value / Shares

C. Substitution and Working:

TV4 = 61×1.05/(0.115−0.05) = 64.05/0.065 = ₹985.38 crore

PV of FCFF (years 1–4, at 11.5%) ≈ ₹150.58 crore

PV of TV4 ≈ ₹637.54 crore

EV ≈ 150.58 + 637.54 = ₹788.11 crore

Equity Value = 788.11 − 150 + 40 = ₹678.11 crore

Value per share = 678.11/20 = ₹33.91

D. Final Result: Enterprise value ≈ ₹788.11 crore; equity value ≈ ₹678.11 crore; value per share ≈ ₹33.91.

E. Interpretation: At a market price of ₹28 per share, the stock trades at roughly a 17.5% discount to the estimated value per share, implying the market may be undervaluing the company's forecast free cash flow growth or applying a more conservative view of achievable margins and reinvestment needs.

F. Decision/Recommendation: Given the meaningful margin of safety, a Buy recommendation is supported, with the ₹33.91 figure serving as the base-case target price, subject to the usual sensitivity around WACC and terminal growth.

G. Teaching Insight: This case is a complete, compact walk-through of the full FCFF-to-equity-value bridge; a common student error is forgetting to add cash (or subtracting it) when moving from enterprise value to equity value, or discounting the terminal value back the wrong number of years.

SECTION IV

Terminal Cash Flow, Terminal Value, and Long-Term Growth

Case 28: Normalizing Terminal-Year Cash Flow

Learning Objective: Avoid using an abnormal final explicit-year cash flow as the base for terminal valuation.

Case Situation

A listed hotel company's fifth year (final explicit forecast year) free cash flow is projected at only ₹60 crore, well below its underlying run-rate, because the company plans a one-time ₹25 crore refurbishment of two properties that year. Beyond year five, cash flow is expected to grow at a stable long-term rate. An analyst preparing the terminal-value calculation must decide whether to grow the reported ₹60 crore figure directly or first adjust it.

Required

72. Estimate the normalized terminal-year cash flow.

73. Explain why simply growing the reported fifth-year cash flow would understate value.

Solution

A. Relevant Valuation Principle: Terminal value should be based on a sustainable, normalized cash flow representative of the business's ongoing run-rate, not a single explicit year distorted by non-recurring items.

B. Formula: Normalized Terminal-Year Cash Flow = Reported Cash Flow + One-Time/Non-Recurring Outflow Add-back

C. Substitution and Working:

Normalised cash flow = 60 + 25 = ₹85 crore

D. Final Result: Normalized terminal-year cash flow = ₹85 crore, versus the reported ₹60 crore.

E. Interpretation: The one-time refurbishment spend depresses year-five cash flow by ₹25 crore; since this expenditure will not recur every year into perpetuity, using the reported ₹60 crore as the terminal-value base would understate the company's true sustainable cash-generating capacity by nearly 42%.

F. Decision/Recommendation: The analyst should use the normalized ₹85 crore figure, not the reported ₹60 crore, as the basis for computing terminal value beyond year five.

G. Teaching Insight: A common error is applying the terminal-growth formula directly to whatever cash flow number appears in the final explicit year, without checking for one-time items. This case is a useful trigger for the habit of scanning the terminal year specifically for non-recurring distortions.

Case 29: Perpetuity-Growth Terminal Value

Learning Objective: Estimate terminal value using sustainable cash flow, long-term growth, and the appropriate discount rate.

Case Situation

A mature household-products company is expected to generate FCFF of ₹180 crore in year five of the explicit forecast, with stable reinvestment needs thereafter consistent with a long-term growth rate of 5%. The company's WACC is 11%. An analyst preparing a full DCF wants the terminal value at the end of year five, its present value, and a sense of how much the total enterprise value arises from beyond the explicit forecast period.

Required

74. Estimate the terminal value at the end of year five.
75. Estimate the present value of the terminal value.
76. Comment on the proportion of total value is likely to arise from the terminal period.

Solution

A. Relevant Valuation Principle: The perpetuity-growth method estimates the terminal value as the present value, at the end of the explicit period, of a cash flow growing at a stable long-term rate in perpetuity thereafter.

B. Formula: $TV5 = FCFF5(1+g) / (WACC - g)$; $PV(TV5) = TV5 \times (1+WACC)^{-5}$

C. Substitution and Working:

$TV5 = 180 \times 1.05 / (0.11 - 0.05) = 189/0.06 =$ ₹3,150.00 crore

$PV(TV5) = 3{,}150.00 \times (1.11)^{-5}$ = ₹1,869.37 crore

D. Final Result: Terminal value at end of year 5 = ₹3,150.00 crore; present value of terminal value ≈ ₹1,869.37 crore.

E. Interpretation: For a mature, low-growth company valued over a modest explicit period, the terminal value typically represents a very large share (often 70–85%) of total enterprise value — this case is no exception, underscoring that even 'stable' companies carry significant sensitivity to the long-term growth and discount-rate assumptions embedded in the terminal calculation.

F. Decision/Recommendation: The analyst should present the ₹1,869.37 crore present value of terminal value alongside a sensitivity table for WACC and terminal growth, given how dominant this single figure is likely to be within total enterprise value.

G. Teaching Insight: Reinforce the standard checks: terminal growth must always be below WACC, and terminal growth should never exceed a reasonable estimate of long-term nominal GDP growth for a mature company operating primarily in a single economy.

Case 30: Exit-Multiple Terminal Value

Learning Objective: Estimate terminal value using an industry multiple and understand the assumptions embedded in that multiple.

Case Situation

An auto-dealership platform is expected to generate EBITDA of ₹120 crore in year five of the explicit forecast. Comparable listed auto-dealership businesses currently trade in a range of 7.0x to 9.0x EV/EBITDA. An analyst has been asked to estimate a terminal value using the midpoint of this range, while also flagging the risk inherent in this approach.

Required

77. Estimate the terminal value using the midpoint exit multiple.
78. Discuss the risk of relying on a current market multiple applied several years into the future.

Solution

A. Relevant Valuation Principle: The exit-multiple method estimates terminal value by applying a current market-observed multiple to a forecast financial metric in the terminal year, implicitly assuming today's market pricing conditions will persist.

B. Formula: TV5 = EBITDA5 × Exit Multiple (midpoint of comparable range)

C. Substitution and Working:

Midpoint multiple = (7.0 + 9.0)/2 = 8.0x

TV5 = 120 × 8.0 = ₹960.00 crore

D. Final Result: Terminal value at end of year 5 = ₹960.00 crore, using an 8.0x EV/EBITDA exit multiple.

E. Interpretation: This approach implicitly assumes that auto-dealership businesses will still command a 7.0x–9.0x multiple five years from now — an assumption that could be undermined by changes in industry structure (for example, electric-vehicle transition, direct-to-consumer sales models, or shifts in capital-market sentiment toward the sector) well before the terminal year arrives.

F. Decision/Recommendation: The ₹960.00 crore estimate can be used as a cross-check against a perpetuity-growth terminal value but should not be treated as more 'objective' simply because it is based on observable market multiples — both methods embed forward-looking assumptions.

G. Teaching Insight: A useful discussion point: students often perceive the exit-multiple method as less assumption-laden than the perpetuity-growth method because it uses 'real' market data. Emphasize that the multiple itself is a snapshot of current sentiment and structural conditions, which may not hold five years out.

Case 31: Long-Term Growth and the Indian Economy

Learning Objective: Assess whether a perpetual growth assumption is economically sustainable.

Case Situation

An analyst valuing a mature Indian building-materials company has used a perpetual (terminal) growth rate of 9% in a DCF model, arguing this reflects the company's strong recent historical growth. A senior colleague reviewing the model points out that India's long-term nominal GDP growth is more realistically estimated at 10–11% and questions whether any single mature company should be assumed to grow faster than, or even in line with, the broader economy forever.

Required

79. Identify the problem with the analyst's 9% perpetual growth assumption.
80. Select a more defensible long-term growth rate from a reasonable range (4%–6%) for a mature company and justify the choice.
81. Explain the valuation implication of revising the growth assumption downward.

Solution

A. Relevant Valuation Principle: A perpetual growth rate should not exceed, and is typically set meaningfully below, the long-term nominal growth rate in which the company operates, since no single mature company can outgrow the aggregate economy forever without eventually representing an implausibly large share of it.

B. Formula: Terminal Value = CF(1+g)/(WACC − g); higher g (holding WACC fixed) mechanically raises terminal value, often substantially, given the sensitivity of the (WACC − g) denominator.

C. Substitution and Working:

A rate near India's long-term nominal GDP growth (9%) is too aggressive for a single mature company to sustain in perpetuity; a defensible range for a mature, low-growth building-materials business is closer to 4%–6%, reflecting real GDP growth plus modest inflation, without assuming indefinite market-share gains.

D. Final Result: A perpetual growth rate of approximately 5% is more defensible than the analyst's original 9% assumption.

E. Interpretation: Because the terminal value is highly sensitive to the (WACC − g) spread, reducing g from 9% to 5% holding WACC fixed — would substantially lower the terminal value and, by extension, the overall enterprise value, likely correcting what was probably a meaningfully overstated valuation.

F. Decision/Recommendation: The analyst should revise the terminal growth assumption down to roughly 4%–6%, rerun the terminal-value calculation, and treat any prior valuation built on the 9% assumption as unreliable until corrected.

G. Teaching Insight: Use this case to instill a standing discipline: terminal growth assumptions should always be benchmarked against a country's or region's long-term nominal GDP growth as a sanity check, not derived purely from a company's recent historical growth trajectory.

Case 32: Transition from High Growth to Stable Growth

Learning Objective: Link stable growth to lower reinvestment, mature margins, payout, and risk.

Case Situation

An Indian digital payments company is expected to mature gradually over a 4-year transition period. Revenue growth is expected to decline from 40% to 12% over the four years, while operating margin improves from 5% to 15% as the business achieves scale, and capital-expenditure-to-revenue intensity falls from 18% to 6% as infrastructure investment needs ease. An analyst preparing the terminal-year cash-flow base must decide how to characterize the terminal-year profile rather than simply applying a lower growth rate to the current cash flow.

Required

82. Describe how revenue growth, margin, and reinvestment intensity should evolve by the end of the transition period.
83. Explain why the terminal-year cash-flow profile must be rebuilt from these evolving drivers rather than derived by applying a single lower growth rate to today's cash flow.

Solution

A. Relevant Valuation Principle: A defensible terminal-year cash flow reflects the entire mature-stage operating profile (margin, reinvestment needs, and payout consistent with lower growth) — not merely today's cash flow multiplied by a smaller growth rate, since a maturing business's cost structure and reinvestment needs change fundamentally, not just its growth rate.

B. Formula: (Conceptual case — terminal-year FCFF should be rebuilt as Revenue terminal × Margin terminal × (1−t), less mature-stage reinvestment, rather than Cash Flow today × (1+g_stable))

C. Substitution and Working:

By the end of the transition period, growth has fallen to 12%, margin has risen to 15%, and capex intensity has fallen to 6% of revenue — a materially different operating profile from today's 40%-growth, 5%-margin, 18%-capex-intensity business. Simply taking today's (low-margin, high-capex) cash flow and growing it at a lower rate would understate the mature-stage cash flow, since it ignores the margin expansion and falling reinvestment intensity that accompany maturation.

D. Final Result: The terminal-year cash flow must be reconstructed using the mature-stage margin (15%) and mature-stage capital intensity (6% of revenue), not extrapolated from today's cash flow.

E. Interpretation: A digital-payments business scaling toward maturity typically sees costs spread over a larger revenue base (margin expansion) and a falling need for infrastructure investment relative to revenue (lower capex intensity) — both of which raise sustainable free cash flow well beyond what a simple lower-growth extrapolation of today's cash flow would suggest.

F. Decision/Recommendation: The analyst should build the terminal year explicitly using the mature-stage margin and capex-intensity assumptions, and only then apply the 12% (declining toward stable) growth rate — not shortcut the process by growing today's depressed cash flow at a lower rate.

G. Teaching Insight: This case addresses a frequent shortcut error: assuming a 'transition period' means only slowing the growth rate, when in a genuinely maturing business, the margin and reinvestment profile change just as materially as the growth rate itself.

Case 33: Terminal-Value Sensitivity

Learning Objective: Evaluate how small changes in WACC and perpetual growth can materially affect intrinsic value.

Case Situation

An analyst has built a base-case DCF for a consumer company using a fifth-year FCFF of ₹180 crore, a WACC of 11%, and a terminal growth rate of 5%, arriving at a base-case terminal value. The investment committee wants to see how this terminal value would change under two alternative WACC-growth combinations: a more conservative case (WACC 12%, growth 4%) and a more optimistic case (WACC 10%, growth 6%), to judge how dependent the recommendation is on these assumptions.

Required

84. Recompute the terminal value under the conservative and optimistic WACC-growth combinations.
85. Compare the range of terminal values with the base case.
86. Comment on whether the investment recommendation is likely to be robust or highly assumption-dependent.

Solution

A. Relevant Valuation Principle: Terminal value is acutely sensitive to the spread between WACC and perpetual growth; even modest, plausible shifts in either input can move the terminal value — and hence the recommendation — substantially.

B. Formula: TV5 = FCFF5(1+g)/(WACC − g), recomputed under each WACC-g combination

C. Substitution and Working:

Base case (WACC 11%, g 5%): TV5 = 180×1.05/0.06 = ₹3,150.00 crore

Conservative (WACC 12%, g 4%): TV5 = 180×1.04/0.08 = ₹2,340.00 crore

Optimistic (WACC 10%, g 6%): TV5 = 180×1.06/0.04 = ₹4,770.00 crore

D. Final Result: Terminal value ranges from ₹2,340.00 crore (conservative) to ₹4,770.00 crore (optimistic), against a base case of ₹3,150.00 crore.

E. Interpretation: A combined one-percentage-point shift in WACC and growth (in opposite directions) moves the terminal value by roughly −26% to +51% relative to the base case — an enormous range for what appear to be modest, individually plausible input changes, driven entirely by how the (WACC − g) denominator compresses or expands.

F. Decision/Recommendation: The investment committee should treat the base-case recommendation as meaningfully assumption-dependent rather than robust and should require the analyst to present the full sensitivity range (not just the base case) before any position-sizing decision is made.

G. Teaching Insight: This case is an effective way to make the abstract idea of 'terminal-value sensitivity' concrete and visual. Encourage students to always test at least a conservative and an optimistic WACC-growth pairing before finalizing any DCF-based recommendation.

SECTION V

P/E Valuation and PEG Analysis

Case 34: Stable-Growth Justified P/E

Learning Objective: Derive a justified P/E from payout, growth, and the required return.

Case Situation

A mature listed FMCG company maintains a return on equity of 20% and a dividend payout ratio of 50%. Its cost of equity is estimated at 15%. The stock currently trades at a P/E of 24x. A research analyst wants to determine the justified leading P/E implied by these fundamentals and compare it with the market multiple.

Required

87. Estimate the sustainable growth rate and the justified leading P/E.
88. Compare the justified P/E with the market P/E of 24x and comment on the implication.

Solution

A. Relevant Valuation Principle: The justified P/E is derived from the payout ratio, sustainable growth, and the required return, providing a fundamentals-based benchmark against which the market-observed multiple can be assessed.

B. Formula: $g = b \times ROE$; Justified $P/E = \text{Payout} \times (1+g) / (K_e - g)$

C. Substitution and Working:

$b = 1 - 0.50 = 0.50$; $g = 0.50 \times 0.20 = 10.0\%$

Justified $P/E = 0.50 \times 1.10 / (0.15 - 0.10) = 0.55/0.05 = 11.00x$

D. Final Result: Justified leading P/E ≈ 11.00x, against a market P/E of 24x.

E. Interpretation: The market multiple (24x) is more than double the fundamentals-based justified multiple (11.00x), suggesting the stock is pricing in either materially higher future growth, a lower risk perception (and hence lower required return) than assumed, or simply a substantial premium for brand strength and earnings quality not captured in this simple model.

F. Decision/Recommendation: The wide gap warrants caution: the analyst should not conclude the stock is overvalued purely on this basis but should treat the justified P/E as one data point requiring reconciliation with qualitative factors (brand moat, capital-light model) before any recommendation.

G. Teaching Insight: A common student error is treating the justified P/E as the 'correct' price and immediately calling the stock overvalued. Emphasize that justified-P/E models are highly simplified and often understate value for high-quality, capital-light franchises — the gap itself is the analytical finding, not an automatic verdict.

Case 35A: Two-Stage P/E Valuation

Learning Objective: Estimate a justified P/E for a company with temporary high growth and changing payout.

Case Situation

An Indian specialty retailer currently has an EPS of ₹10.00 and is expected to grow earnings at 25% annually for the next five years while paying out only 10% of earnings during this expansion. Thereafter, growth is expected to stabilize at 8%, supported by a sustainable ROE of 18%, with payout rising to a level consistent with that stable growth. The cost of equity is 14%. The stock currently trades at 20x trailing earnings.

Required

89. Estimate the EPS and dividend for each of the five high-growth years.
90. Estimate the stable-stage payout, terminal value, and the implied two-stage justified P/E.
91. Assess whether the prevailing market multiple of 20x is justified.

Solution

A. Relevant Valuation Principle: A two-stage P/E valuation values the explicit high-growth dividend stream separately from a terminal value based on stable-stage payout, growth, and ROE, then expresses the resulting intrinsic value as an implied multiple of current EPS.

B. Formula: Dt = EPSt × payout_hg; stable b = g_stable/ROE_stable; TV5 = D6/(Ke − g_stable); Implied P/E = V0/EPS0

C. Substitution and Working:

EPS (years 1–5): ₹12.50, 15.63, 19.53, 24.41, 30.52;

Dividends (10% payout): ₹1.25, 1.56, 1.95, 2.44, 3.05

Stable payout = 1 − (8%/18%) = 55.56%; EPS6 = 30.52×1.08 = ₹32.96; D6 = ₹18.31

TV5 = 18.31/(0.14−0.08) = ₹305.18; PV of dividends ≈ ₹7.65; PV of TV5 ≈ ₹158.50

V0 ≈ 7.65+158.50 = ₹166.15; Implied P/E = 166.15/10.00 ≈ 16.51x

D. Final Result: Implied two-stage justified P/E ≈ 16.51x, against a market P/E of 20x.

E. Interpretation: The market multiple of 20x sits meaningfully above the fundamentals-based justified multiple of 16.51x, suggesting the current price embeds either a longer or steeper high-growth period than modeled here, or a lower perceived risk — the retailer's aggressive 25% growth assumption already looks generous, so the market appears to be pricing in an even more optimistic scenario.

F. Decision/Recommendation: Given the market multiple already exceeds a fairly optimistic justified P/E, the analyst should treat the stock as, at best, fully valued, and should avoid recommending fresh purchases at the prevailing 20x multiple without a more aggressive (and less easily defensible) growth assumption.

G. Teaching Insight: This case shows that even a generous high-growth assumption (25% for five years) does not always justify the prevailing market multiple — students should resist the temptation to keep raising growth or extending its duration merely to make the model match the market price.

Case 35B: Two-Stage P/E Valuation — Renewable Energy (Solar) Company

Learning Objective: Estimate a justified P/E when the high-growth phase is short but exceptionally rapid, and observe how quickly value becomes almost entirely dependent on terminal-stage assumptions when both the growth window is brief and payout is minimal.

Case Situation
Suryodaya Renewable Power Ltd, an Indian solar power developer, currently has an EPS of ₹8.00 and is expected to grow earnings at 35% annually for the next three years as it commissions a pipeline of new solar capacity, paying out only 5% of earnings during this phase as nearly all cash is deployed into new projects. Thereafter, growth is expected to stabilize at 7%, supported by a sustainable ROE of 15% once the portfolio matures into steady, utility-like cash flows. The cost of equity is 15%, reflecting the sector's project execution and regulatory risks. The stock currently trades at 18x trailing earnings.

Required

1. Estimate the EPS and dividend for each of the three high-growth years.
2. Estimate the stable-stage payout, terminal value, and the implied two-stage justified P/E.
3. Assess whether the prevailing market multiple of 18x is justified.

Solution

A. Relevant Valuation Principle: A two-stage P/E valuation values the explicit high-growth dividend stream separately from a terminal value based on stable-stage payout, growth, and ROE, then expresses the resulting intrinsic value as an implied multiple of current EPS.

B. Formula: Dt = EPSt × payout_hg; stable b = g_stable/ROE_stable; TV3 = D4/(Ke − g_stable);
Implied P/E = V0/EPS0

C. Substitution and Working:
EPS (years 1–3): ₹10.80, 14.58, 19.68
Dividends (5% payout): ₹0.54, 0.73, 0.98
PV of each dividend (at 15%): ₹0.47, 0.55, 0.65
PV of high-growth-phase dividends = ₹1.67
Stable payout = 1 − (7%/15%) = 53.33%; EPS4 = 19.68×1.07 = ₹21.06; D4 = ₹11.23
TV3 = 11.23/(0.15−0.07) = ₹140.41; PV of TV3 = ₹92.32
V0 = 1.67+92.32 = ₹93.99; Implied P/E = 93.99/8.00 ≈ 11.75x

D. Final Result: Implied two-stage justified P/E ≈ 11.75x, against a market P/E of 18x.

E. Interpretation: With only three years of high growth at a minimal 5% payout, the terminal value carries over 98% of total worth, which is even more extreme than Case 35A's already-terminal-heavy 96%. The market's 18x multiple is roughly 35% above the justified figure, a wider gap than 35A's — for a young, capex-intensive sector, this signals the market may be pricing in either a longer runway of 35% growth than three years, faster-than-modeled capacity additions, or is discounting execution and regulatory risk less heavily than the 15% Ke assumed here.

F. Decision/Recommendation: Given the market multiple sits well above a justified figure that is itself sensitive to just a handful of terminal-stage assumptions, the analyst should treat 18x as pricing in considerable optimism and should stress-test the stable-stage ROE and growth assumptions specifically — in a sector this reliant on terminal value, small changes in those two inputs alone can close much of the gap.

G. Teaching Insight: This case sharpens the terminal-value-dominance lesson from Case 35A by shortening the explicit window from five years to three and thinning the payout further — students should notice that a *shorter* high-growth phase does not mean less exposure to terminal-value risk; if anything, it means the analysis rests on the stable-stage assumptions even more heavily, that much sooner.

Case 35C: Two-Stage P/E Valuation — IT Services Company

Learning Objective: Estimate a justified P/E for a steady, longer-duration grower that pays a meaningful dividend even during its growth phase and see how a real payout sustained over a longer explicit window shifts the balance of value away from the terminal stage.

Case Situation
Vistaar Technologies Ltd, an Indian IT services company, currently has an EPS of ₹15.00 and is expected to grow earnings at a steadier 15% annually for the next six years as it scales its digital-services practice, paying out a meaningful 30% of earnings during this phase given the business's asset-light, cash-generative nature. Thereafter, growth is expected to stabilize at 6%, supported by a sustainable ROE of 20%, with payout rising further to a level consistent with that stable growth. The cost of equity is 13%, reflecting the company's scale and earnings predictability. The stock currently trades at 22x trailing earnings.

Required

1. Estimate the EPS and dividend for each of the six high-growth years.
2. Estimate the stable-stage payout, terminal value, and the implied two-stage justified P/E.
3. Assess whether the prevailing market multiple of 22x is justified.

Solution

A. Relevant Valuation Principle: A two-stage P/E valuation values the explicit high-growth dividend stream separately from a terminal value based on stable-stage payout, growth, and ROE, then expresses the resulting intrinsic value as an implied multiple of current EPS.

B. Formula: Dt = EPSt × payout_hg; stable b = g_stable/ROE_stable; TV6 = D7/(Ke − g_stable); Implied P/E = V0/EPS0

C. Substitution and Working:
EPS (years 1–6): ₹17.25, 19.84, 22.81, 26.24, 30.17, 34.70
Dividends (30% payout): ₹5.17, 5.95, 6.84, 7.87, 9.05, 10.41
PV of each dividend (at 13%): ₹4.58, 4.66, 4.74, 4.83, 4.91, 5.00
PV of high-growth-phase dividends = ₹28.72
Stable payout = 1 − (6%/20%) = 70.0%; EPS7 = 34.70×1.06 = ₹36.78; D7 = ₹25.74

TV6 = 25.74/(0.13−0.06) = ₹367.78; PV of TV6 = ₹176.65
V0 = 28.72+176.65 = ₹205.37; Implied P/E = 205.37/15.00 ≈ 13.69x

D. Final Result: Implied two-stage justified P/E ≈ 13.69x, against a market P/E of 22x.

E. Interpretation: Here, the explicit high-growth phase contributes a much more substantial 14% of total value — a sharp contrast to 35A (4%) and 35B (under 2%) — because a longer six-year window combined with a real 30% payout means meaningful cash reaches shareholders well before the terminal stage. Even so, the market's 22x multiple sits about 38% above the justified 13.69x, a wider proportional gap than either 35A or 35B, despite this being the most "grounded" of the three cases in terms of how much of its value is verifiable in the near term.

F. Decision/Recommendation: Because a larger share of this valuation rests on the explicit, more forecastable six-year period rather than the terminal stage alone, the analyst has comparatively more confidence in the 13.69x figure than in 35A or 35B's justified multiples, which makes the 38% gap to the market price harder to explain away as terminal-assumption uncertainty. The stock should be treated as significantly overvalued at 22x, with the burden of proof on anyone arguing that growth will run meaningfully longer than six years.

G. Teaching Insight: Read alongside 35A and 35B, this case completes a three-way spectrum: a short, thin-payout, terminal-dominated story (35B, 98% terminal); a moderate story (35A, 96% terminal); and a longer, real-payout story where the explicit period finally contributes a meaningful share (35C, 86% terminal). Students should take from this progression that terminal-value dominance is not an inherent flaw of the two-stage model — it is a direct, quantifiable consequence of how short the growth window is and how little of it is paid out along the way.

Case 36: Retention Ratio, ROE, and P/E

Learning Objective: Demonstrate that increasing retention does not necessarily increase the justified P/E.

Case Situation

Three hypothetical Indian companies operate with the same cost of equity of 14% and the same 60% earnings retention ratio but differ in ROE: Company X earns an ROE of 10% (below its cost of equity), Company Y earns exactly 14% (equal to its cost of equity), and Company Z earns 20% (above its cost of equity). An investment analyst wants to examine, without lengthy calculation, how retention affects the justified P/E differently across the three companies.

Required

92. Estimate the sustainable growth rate for each company under the common 60% retention policy.
93. Rank the three companies by their likely justified P/E, without necessarily computing a precise figure for each.
94. Explain which company should retain more earnings, and which should not.

Solution

A. Relevant Valuation Principle: Retention only raises the justified P/E when ROE exceeds the cost of equity; when ROE is below Ke, higher retention actually lowers value, and when ROE equals Ke, retention has no effect on value (all growth is value-neutral).

B. Formula: g = b × ROE; direction of P/E impact of retention depends on the sign of (ROE − Ke)

C. Substitution and Working:

Company X (ROE 10% < Ke 14%): g = 6.0%, but reinvestment earns below the required return, so higher retention would reduce the justified P/E relative to a lower-retention policy.

Company Y (ROE 14% = Ke 14%): g = 8.4%, but reinvestment exactly matches the required return, so the level of retention is value-neutral — the justified P/E is the same regardless of payout policy.

Company Z (ROE 20% > Ke 14%): g = 12.0%, and reinvestment earns well above the required return, so higher retention meaningfully raises the justified P/E.

D. Final Result: Ranking of justified P/E (highest to lowest, under the same 60% retention policy): Company Z > Company Y > Company X.

E. Interpretation: The identical 60% retention policy produces value-creating growth only for Company Z; for Company Y, it is value-neutral, and for Company X, it is mildly value-destroying — illustrating that the level of retention alone tells you nothing about value creation without knowing where ROE sits relative to Ke.

F. Decision/Recommendation: Company Z should retain more earnings (increase retention further, if attractive reinvestment opportunities exist), Company Y is indifferent to its retention level from a pure value standpoint, and Company X should reduce retention (raise payout) rather than reinvest at a sub-par return.

G. Teaching Insight: This case is best used as a quick, discussion-based exercise rather than a heavy numerical one — the qualitative insight (retention's value impact depends entirely on the ROE-Ke spread) is the actual learning objective, not the arithmetic.

Case 37: Trailing versus Forward P/E

Learning Objective: Select the earnings measure that best represents expected performance.

Case Situation

A cyclical steel company is recovering from a weak year in which plant utilization was depressed by a temporary raw-material shortage. Trailing twelve-month EPS is only ₹5.00, reflecting this weak year, while next year's expected EPS, based on normal utilization levels, is ₹15.00. The stock currently trades at ₹150 per share. An analyst must decide which P/E ratio better represents the company's investment merit.

Required

95. Calculate the trailing P/E and the forward P/E.
96. Decide which measure is more informative for this company, with appropriate caution.

Solution

A. Relevant Valuation Principle: For cyclical companies recovering from an abnormal period, trailing earnings can badly distort the P/E ratio; forward earnings, based on a more normal operating environment, are usually more informative — but rely on a forecast rather than reported fact.

B. Formula: Trailing P/E = CMP / Trailing EPS; Forward P/E = CMP / Forward EPS

C. Substitution and Working:

Trailing P/E = 150/5.00 = 30.00x

Forward P/E = 150/15.00 = 10.00x

D. Final Result: Trailing P/E = 30.00x; Forward P/E = 10.00x.

E. Interpretation: The trailing P/E of 30.00x looks expensive purely because it is calculated on a depressed, abnormal earnings base; the forward P/E of 10.00x, based on normalized utilization, is a much more meaningful reflection of the stock's valuation relative to its expected earning power — but forward EPS is itself a forecast and carries its own execution risk (utilization may not normalize as expected).

F. Decision/Recommendation: The analyst should base the investment view primarily on the forward P/E of 10.00x, while explicitly flagging the forecast risk around next year's expected utilization recovery, rather than relying on the misleadingly high trailing multiple.

G. Teaching Insight: This case is a good reminder that a high trailing P/E is not always a valuation red flag — it can simply reflect a depressed earnings base for a cyclical business. Encourage students to always check whether reported earnings reflect a normal operating environment before interpreting any P/E ratio.

Case 38: Negative Earnings and the Limitation of P/E

Learning Objective: Recognize when the P/E ratio is mathematically or economically meaningless.

Case Situation

A listed food-delivery platform reported a net loss for the year, yet revenue grew 45%, and contribution margin (revenue less delivery and platform costs) improved meaningfully as order density increased across cities. An equity research trainee, following standard practice, has attempted to calculate the company's P/E ratio and is confused by the negative result.

Required

97. Explain why the P/E ratio cannot be meaningfully used for this company.
98. Recommend a more appropriate relative-valuation measure given the available data.

Solution

A. Relevant Valuation Principle: The P/E ratio is undefined or economically meaningless when earnings are negative, since a negative denominator produces a nonsensical (negative or mathematically inconsistent) multiple that cannot be compared across companies or interpreted as a valuation signal.

B. Formula: (Conceptual case — no P/E calculation is meaningful with negative EPS.)

C. Substitution and Working:

With EPS negative, P/E = Price/EPS produces a negative number that has no economic interpretation as a valuation multiple (a negative P/E does not mean 'cheap'); the ratio should simply be treated as not applicable rather than computed and reported.

D. Final Result: P/E is not applicable for this company in its current loss-making state.

E. Interpretation: Since the company has observable, rapidly growing revenue and an improving contribution margin, a revenue-based multiple (P/S) or a unit-economics-based measure (value per order, or EV/Gross Merchandise Value) is more appropriate, as these do not require positive earnings and can still capture the improving underlying economics.

F. Decision/Recommendation: The trainee should switch to a P/S-based (or unit-economics-based) valuation framework for this company, clearly noting that P/E will only become meaningful once the business achieves sustained profitability.

G. Teaching Insight: A common student instinct is to force-fit a P/E ratio regardless of the sign of earnings. This case reinforces the discipline of checking whether a valuation multiple's underlying denominator is economically meaningful before using it, rather than mechanically applying a formula.

Case 39: PEG Ratio for a Growth Company

Learning Objective: Use PEG as a supplementary rather than mechanical valuation rule.

Case Situation

Two Indian technology services companies are being compared by a research analyst. Company A trades at a P/E of 25x and is expected to grow earnings at 20% annually. Company B trades at a lower P/E of 15x but is expected to grow earnings at only 8% annually. A junior colleague argues that PEG ratios will settle the question of which stock is cheaper.

Required

99. Calculate the PEG ratio for both companies.
100. Identify which stock appears cheaper on a PEG basis.
101. Discuss why risk, growth duration, and return on capital must also be considered before relying on PEG alone.

Solution

A. Relevant Valuation Principle: The PEG ratio (P/E divided by expected growth rate) is a useful supplementary screen for comparing growth-adjusted valuation, but it implicitly assumes growth of equal quality, duration, and risk across companies — an assumption that rarely holds in practice.

B. Formula: PEG = P/E ÷ Expected Growth Rate (expressed as a whole number, not a decimal)

C. Substitution and Working:

Company A: PEG = 25/20 = 1.25

Company B: PEG = 15/8 = 1.875

D. Final Result: PEG: Company A = 1.25; Company B = 1.875.

E. Interpretation: On a PEG basis, Company A appears cheaper (lower PEG) despite its higher headline P/E, because its growth rate is proportionately higher. However, PEG treats a percentage-point of growth as equally valuable and equally certain for both companies — it does not account for how long each company can sustain its growth rate, how much capital is required to generate it, or the relative risk that each company's growth assumptions will materialize.

F. Decision/Recommendation: The analyst should use the PEG comparison (favoring Company A) only as a starting point and should specifically investigate the sustainability and capital intensity of each company's growth — including return on capital employed — before drawing a firm conclusion about relative cheapness.

G. Teaching Insight: This case should be used to caution against treating PEG as a definitive screen. A lower PEG does not automatically mean 'cheaper' in any risk-adjusted sense — encourage students to ask what is driving each company's growth rate before trusting the ratio.

Case 40: Peer P/E and Investment Recommendation

Learning Objective: Apply peer multiples while adjusting for differences in growth, risk, and earnings quality.

Case Situation

A mid-cap pharmaceutical company has an EPS of ₹12.00 and trades at a P/E of 16x, while its listed peer group has a median P/E of 22x. However, the company faces meaningful customer concentration risk (a small number of clients account for a large share of revenue) and some regulatory uncertainty regarding a key manufacturing facility. An analyst must decide whether the full peer discount is a buying opportunity or is justified by company-specific risk.

Required

102. Calculate the current discount to the peer median P/E.
103. Estimate a target price using a reasonably adjusted multiple, rather than the full peer median.
104. Explain why the full peer median multiple may not be justified for this company.

Solution

A. Relevant Valuation Principle: Peer multiples must be adjusted downward for company-specific risks (concentration, regulatory uncertainty) that are not present, or are less severe, across the broader peer set — applying the unadjusted peer median would overstate fair value.

B. Formula: Discount (%) = (Peer Median P/E − Company P/E)/Peer Median P/E; Target Price = Adjusted Multiple × EPS

C. Substitution and Working:

Discount = (22 − 16)/22 = 27.27%

Using an adjusted multiple of 19x (between the company's current 16x and the peer median 22x, reflecting partial but not full risk compression): Target price = 19 × 12.00 = ₹228.00

Current market price = 16 × 12.00 = ₹192.00

D. Final Result: The company trades at a 27.27% discount to the peer median; using an adjusted multiple of 19x, target price ≈ ₹228.00 versus current price of ₹192.00 (about 18.75% upside).

E. Interpretation: While the stock trades well below the peer median, applying the full 22x peer multiple would ignore genuine, company-specific risks (customer concentration, regulatory uncertainty at a key facility) that peers may not share to the same degree; a partial re-rating toward 19x, rather than the full peer median, more fairly balances the discount the market is applying for these real risks against the possibility that some of the discount is excessive.

F. Decision/Recommendation: A Buy recommendation is supported, targeting ₹228.00 (19x adjusted multiple) rather than the full peer-median-implied ₹264.00 (22x), explicitly acknowledging that customer concentration and regulatory risk justify some, but not all, of the current discount.

G. Teaching Insight: This case addresses a common analytical shortcut: mechanically applying the peer median multiple to any 'cheap looking' stock. Reinforce that peer comparisons require judgment about which company-specific risks justify part of an observed discount, and which do not.

SECTION VI

P/B Valuation

Case 41: Stable-Growth Justified P/B

Learning Objective: Estimate P/B using ROE, payout, growth, and required return.

Case Situation

A mature housing-finance company maintains a stable return on equity of 16% and a dividend payout ratio of 30%. Its cost of equity is estimated at 13%. The stock currently trades at a P/B multiple of 3.2x. An analyst wants to determine the justified P/B implied by these fundamentals and compare it with the observed market multiple.

Required

105. Estimate the sustainable growth rate and the justified P/B.
106. Compare the justified P/B with the observed market P/B of 3.2x and comment on the implications.

Solution

A. Relevant Valuation Principle: The justified P/B multiple links ROE, sustainable growth, and the required return, and rises above 1.0x precisely to the extent that ROE exceeds the cost of equity.

B. Formula: $g = b \times ROE$; Justified P/B = $(ROE - g) / (Ke - g)$

C. Substitution and Working:

$b = 1 - 0.30 = 0.70$; $g = 0.70 \times 0.16 = 11.2\%$

Justified P/B = $(0.16 - 0.112)/(0.13 - 0.112) = 0.048/0.018 = 2.67x$

D. Final Result: Justified P/B ≈ 2.67x, against an observed market P/B of 3.2x.

E. Interpretation: The stock trades at a modest premium (about 20%) to its fundamentals-based justified P/B, suggesting the market may be pricing in a somewhat higher sustainable ROE or lower risk than the current inputs assume, though the gap is not so large as to signal a clear overvaluation on its own.

F. Decision/Recommendation: The analyst can treat the stock as modestly expensive relative to its own fundamentals, warranting a Hold rather than a fresh Buy, pending confirmation of whether ROE is likely to improve further.

G. Teaching Insight: A common student error is confusing the justified P/B formula with the justified P/E formula or forgetting that the numerator uses (ROE − g), not simply ROE. Reinforce the derivation from the residual-income logic underlying the P/B model.

Case 42A: Two-Stage P/B Valuation — Private-Sector Lender (zero growth-phase payout)

Learning Objective: Value a company whose high ROE and growth are expected to decline to stable levels.

Case Situation
An Indian private-sector lender currently has a book value of ₹100 per share and an ROE of 22%, which is expected to persist for five years as the bank fully retains earnings to fund rapid loan-book growth (zero payout during this phase). Thereafter, ROE is expected to normalize to 15% with a stable payout ratio of 40%. The cost of equity is 14%. An analyst wants the two-stage justified P/B implied by this profile.

Required

1. Project the book value per share at the end of the five-year high-growth phase.
2. Estimate the present value of any dividends paid during the high-growth phase, and the terminal-stage justified P/B and value at year five.
3. Estimate the implied two-stage justified P/B as of today.

Solution

A. Relevant Valuation Principle: The complete two-stage P/B has two components: the present value of dividends paid during the high-growth phase, plus the present value of a stable-stage terminal multiple applied to book value at the end of that phase. Each year's dividend is computed on that year's own book value.

B. Formula: $BV_t = BV_0 \times (1+g\_hg)^t$, where $g\_hg=(1-payout\_hg)\times ROE\_hg$; $D_t = payout\_hg \times ROE\_hg \times BV_t$; Stable-stage Justified P/B = (ROE_st−g_st)/(Ke−g_st); Value5 = BV5×Justified P/B; $P0 = \Sigma PV(D_t) + PV(Value5)$

C. Substitution and Working:
Retention during growth phase = 100%; g_hg = 1.00×0.22 = 22.0%
BV (years 1–5): 122.00, 148.84, 181.58, 221.53, 270.27
Dividends, $D_t = 0.00 \times 0.22 \times BV_t$ (years 1–5): ₹0, 0, 0, 0, 0 *(payout_hg = 0%, so this line is zero regardless of which book value it's paired with)*
PV of high-growth-phase dividends = ₹0.00
Stable-stage: b = 60%, g = 0.60×0.15 = 9.0%; Justified P/B (stable) = (0.15−0.09)/(0.14−0.09) = 1.20x
Value5 = 270.27 × 1.20 = ₹324.32; PV of Value5 = 324.32 × $(1.14)^{-5}$ = ₹168.44
P0 = 0.00 + 168.44 = ₹168.44; Implied two-stage justified P/B = 168.44/100 = **1.68x**

D. Final Result: Book value grows to ₹270.27 per share by year 5; implied two-stage justified P/B as of today ≈ 1.68x.

E. Interpretation: With zero payout during the growth phase, the entire justified value comes from the terminal component. This is the one case of the three where the choice between same-year and lagged book value for the dividend calculation is irrelevant — because the dividend itself is zero either way.

F. Decision/Recommendation: The analyst can use 1.68x as the fundamentals-based justified P/B benchmark for the lender today, to be compared against its current market P/B multiple before forming a recommendation.

G. Teaching Insight: High current ROE alone does not translate one-for-one into a high present-day justified P/B — discounting and the eventual stable-stage multiple both matter.

Case 42B: Two-Stage P/B Valuation — Housing Finance Company (20% growth-phase payout)

Learning Objective: Value a company whose high ROE and growth decline to stable levels, where the high-growth phase itself pays a modest dividend computed on that year's own book value.

Case Situation
Griha Nirman Housing Finance Ltd, an Indian housing-finance company, currently has a book value of ₹150 per share and an ROE of 20%, expected to persist for four years as the company grows its mortgage book — but unlike a pure-retention growth story, the company maintains a modest 20% payout even during this phase to satisfy income-seeking shareholders. Thereafter, ROE is expected to normalize to 14% with a stable payout ratio of 50%. The cost of equity is 13%. An analyst wants the two-stage justified P/B implied by this profile.

Required

1. Project the book value per share at the end of the four-year high-growth phase, accounting for the 20% payout.
2. Estimate the present value of the dividends paid during the high-growth phase, and the terminal-stage justified P/B and value at year four.
3. Estimate the implied two-stage justified P/B as of today.

Solution

A. Relevant Valuation Principle: The complete two-stage P/B has two components: the present value of dividends paid during the high-growth phase, plus the present value of a stable-stage terminal multiple applied to book value at the end of that phase. Each year's dividend is computed on that year's own (already-grown) book value, not the prior year's.

B. Formula: $BV_t = BV_0 \times (1+g\_hg)^t$, where g_hg=(1−payout_hg)×ROE_hg; D_t = payout_hg×ROE_hg×BV_t; Stable-stage Justified P/B = (ROE_st−g_st)/(Ke−g_st); Value4 = BV4×Justified P/B; P0 = ΣPV(D_t) + PV(Value4)

C. Substitution and Working:
Retention during growth phase = 1−0.20 = 80%; g_hg = 0.80×0.20 = 16.0%
BV (years 1–4): 174.00, 201.84, 234.13, 271.60
Dividends, $D_t = 0.20 \times 0.20 \times BV_t$ (years 1–4): ₹6.96, 8.07, 9.37, 10.86
PV of each (at 13%): ₹6.16, 6.32, 6.49, 6.66
PV of high-growth-phase dividends = ₹25.64
Stable-stage: b = 50%, g = 0.50×0.14 = 7.0%; Justified P/B (stable) = (0.14−0.07)/(0.13−0.07) = 1.17x

Value4 = 271.60 × 1.17 = ₹316.86; PV of Value4 = 316.86 × $(1.13)^{-4}$ = ₹194.34
P0 = 25.64 + 194.34 = ₹219.97; Implied two-stage justified P/B = 219.97/150 = **1.47x**

D. Final Result: Book value grows to ₹271.60 per share by year 4; the high-growth phase separately contributes ₹25.64 of present value through its dividends; implied two-stage justified P/B as of today ≈ 1.47x.

E. Interpretation: Because each dividend is now paired with that year's own (already-grown) book value rather than the smaller prior-year figure, every dividend in the stream is slightly larger than under a lagged-book-value convention — the interim-dividend component rises to ₹25.64 (about 12% of total value), pulling the overall justified P/B up to 1.47x.

F. Decision/Recommendation: The analyst should use 1.47x, built from both the interim-dividend and terminal components on a consistent same-year book-value basis, as the benchmark.

G. Teaching Insight: Once a growth-phase payout is introduced, students must fix two things at once: include the interim-dividend term at all, and apply it consistently to the same year's book value throughout the projection — not a mix of current- and prior-year figures.

Case 42C: Two-Stage P/B Valuation — Small Finance Bank (10% growth-phase payout)

Learning Objective: Value a company with a very high growth-phase ROE and a modest payout over a longer six-year window, applying the dividend consistently to each year's own book value.

Case Situation
Uday Small Finance Bank Ltd, a recently listed Indian small finance bank, currently has a book value of ₹80 per share and an ROE of 25%, expected to persist for six years as it scales up its retail deposit and lending franchise, with a 10% payout maintained throughout this phase to comply with minimum distribution norms. Thereafter, ROE is expected to normalize to 16% with a stable payout ratio of 45%. The cost of equity is 15%. An analyst wants the two-stage justified P/B implied by this profile.

Required

1. Project the book value per share at the end of the six-year high-growth phase, accounting for the 10% payout.
2. Estimate the present value of the dividends paid during the high-growth phase, and the terminal-stage justified P/B and value at year six.
3. Estimate the implied two-stage justified P/B as of today.

Solution

A. Relevant Valuation Principle: The complete two-stage P/B has two components: the present value of dividends paid during the high-growth phase, plus the present value of a stable-stage terminal multiple applied to book value at the end of that phase. Each year's dividend is computed on that year's own book value.

B. Formula: $BV_t = BV_0 \times (1+g\_hg)^t$, where $g\_hg=(1-payout\_hg)\times ROE\_hg$; $D_t = payout\_hg \times ROE\_hg \times BV_t$; Stable-stage Justified P/B = (ROE_st−g_st)/(Ke−g_st); Value6 = BV6×Justified P/B; P0 = ΣPV(D_t) + PV(Value6)

C. Substitution and Working:
Retention during growth phase = 1−0.10 = 90%; g_hg = 0.90×0.25 = 22.5%
BV (years 1–6): 98.00, 120.05, 147.06, 180.15, 220.68, 270.34
Dividends, $D_t = 0.10\times0.25\times BV_t$ (years 1–6): ₹2.45, 3.00, 3.68, 4.50, 5.52, 6.76
PV of each (at 15%): ₹2.13, 2.27, 2.42, 2.58, 2.74, 2.92
PV of high-growth-phase dividends = ₹15.06
Stable-stage: b = 55%, g = 0.55×0.16 = 8.8%; Justified P/B (stable) = (0.16−0.088)/(0.15−0.088) = 1.16x
Value6 = 270.34 × 1.16 = ₹313.94; PV of Value6 = 313.94 × $(1.15)^{-6}$ = ₹135.73
P0 = 15.06 + 135.73 = ₹150.78; Implied two-stage justified P/B = 150.78/80 = **1.88x**

D. Final Result: Book value grows to ₹270.34 per share by year 6; the high-growth phase separately contributes ₹15.06 of present value through its dividends; implied two-stage justified P/B as of today ≈ 1.88x.

E. Interpretation: As in 42B, pairing each dividend with its own year's book value (rather than the smaller prior-year figure) raises every term in the interim-dividend stream, lifting the justified P/B from what a lagged-book-value convention would have produced (1.85x) to 1.88x.

F. Decision/Recommendation: The analyst should use 1.88x as the benchmark, built consistently on same-year book value throughout.

G. Teaching Insight: Read together, 42A–42C now show a clean progression: zero payout (42A) collapses the model to its terminal-only special case; any positive payout (42B, 42C) activates the interim-dividend term, computed consistently on each year's own book value — and the size of that term scales with both the payout rate and the number of high-growth years it is paid over.

Case 43: ROE-Cost of Equity Spread

Learning Objective: Interpret whether a company should trade above, near, or below book value.

Case Situation

Three Indian financial-services firms operate with the same cost of equity of 14% and are all expected to grow at 5% annually but differ in ROE: Firm P earns 18% (above its cost of equity), Firm Q earns exactly 14% (equal to its cost of equity), and Firm R earns 10% (below its cost of equity). A analyst wants to rank their expected P/B ratios without lengthy calculation, based purely on the ROE-Ke relationship.

Required

107. Rank the three firms by their expected P/B ratio.
108. Explain the value-creation logic behind the ranking without extended computation.

Solution

A. Relevant Valuation Principle: A firm should trade above book value (P/B > 1) when ROE exceeds Ke, at book value (P/B ≈ 1) when ROE equals Ke, and below book value (P/B < 1) when ROE is below Ke — the ROE-Ke spread, not ROE in isolation, determines whether book value understates or overstates a company's worth.

B. Formula: Justified P/B = (ROE − g)/(Ke − g); sign of (ROE − Ke) determines whether P/B is above or below 1.0x

C. Substitution and Working:

Firm P: (0.18−0.05)/(0.14−0.05) = 0.13/0.09 = 1.44x (above 1.0x, since ROE > Ke)

Firm Q: (0.14−0.05)/(0.14−0.05) = 1.00x (exactly at book value, since ROE = Ke)

Firm R: (0.10−0.05)/(0.14−0.05) = 0.05/0.09 = 0.56x (below 1.0x, since ROE < Ke)

D. Final Result: Ranking of expected P/B (highest to lowest): Firm P (1.44x) > Firm Q (1.00x) > Firm R (0.56x).

E. Interpretation: Firm P creates value on every rupee of equity capital employed (earning above what shareholders require), so the market should reasonably price it above book value; Firm Q merely earns what shareholders require, justifying a price at book value; Firm R earns below what shareholders require, so it should logically trade below book value, since incremental equity capital deployed there is not fully compensated.

F. Decision/Recommendation: An investor comparing these three firms purely on relative attractiveness (all else equal) should favour Firm P, treat Firm Q as fairly valued only at book value, and be cautious of Firm R trading anywhere near or above book value, since that would imply the market is not appropriately discounting its sub-par returns.

G. Teaching Insight: This is best run as a quick conceptual exercise to build intuition for the ROE-Ke-P/B relationship before students encounter more complex two-stage P/B problems. The precise multiples matter less than the directional logic.

Case 44: P/B Valuation of a Bank

Learning Objective: Apply P/B in a sector where book equity is an important operating resource.

Case Situation

A fictional scheduled commercial bank has a book value per share of ₹180, a sustainable ROE of 15%, expected growth of 7%, and a cost of equity of 13%. An equity analyst wants to estimate the justified P/B and target price, while also considering whether the bank's asset-quality profile warrants a valuation discount relative to this fundamentals-based figure.

Required

109. Estimate the justified P/B and target price.
110. Comment on whether asset-quality risk should warrant a discount to this target price.

Solution

A. Relevant Valuation Principle: For banks, where book equity directly funds the loan book (an operating resource, not merely a residual accounting figure), the justified P/B framework is a particularly natural valuation tool, though it must be adjusted for asset-quality risk not captured in reported ROE.

B. Formula: Justified P/B = (ROE − g)/(Ke − g); Target Price = Justified P/B × Book Value per Share

C. Substitution and Working:

Justified P/B = (0.15 − 0.07)/(0.13 − 0.07) = 0.08/0.06 = 1.33x

Target price = 1.33 × 180 = ₹240.00

D. Final Result: Justified P/B ≈ 1.33x; target price ≈ ₹240.00.

E. Interpretation: This target price assumes the reported 15% ROE is sustainable and fully reflects asset quality; if the bank's loan book carries elevated stress (higher-than-peer non-performing assets or under-provisioning), the sustainable ROE embedded in this calculation may be overstated, warranting a discount to the ₹240.00 figure.

F. Decision/Recommendation: The analyst should apply a discount to the ₹240.00 target — for example, using a justified P/B closer to 1.1x–1.2x rather than 1.33x — if asset-quality metrics (gross/net NPA ratios, provision coverage) suggest the 15% ROE is not fully risk-adjusted.

G. Teaching Insight: This case highlights that the P/B framework, while well-suited to banks, is only as reliable as the ROE input; for financial institutions, ROE can be temporarily inflated by under-provisioning for stressed assets, making a qualitative asset-quality check an essential companion to the mechanical calculation.

Case 45: Accounting Limitations of P/B

Learning Objective: Recognise how accounting policies and intangible assets reduce the comparability of book values.

Case Situation

An analyst is comparing an asset-heavy cement producer, whose book value largely consists of plant, machinery, and land carried at historical cost less accumulated depreciation, with an IT-consulting company, whose book value is small because its key assets — skilled people, client relationships, and proprietary methodologies — are internally generated and therefore not capitalised on the balance sheet under Indian accounting standards.

Required

111. Explain why P/B is likely to be more meaningful for the cement producer than for the IT-consulting company.
112. Identify the specific effect of internally generated intangibles and historical-cost depreciation on the comparability of the two companies' P/B ratios.

Solution

A. Relevant Valuation Principle: P/B is most meaningful when a company's book value closely approximates its economic asset base; it becomes less reliable when significant value-generating assets (brand, know-how, client relationships) are internally generated and excluded from the balance sheet under accounting rules.

B. Formula: (Conceptual case — no numerical calculation required.)

C. Substitution and Working:

For the cement producer, plant and machinery (tangible, capitalised assets) are the primary value driver, and book value — even adjusted for historical-cost depreciation — reasonably tracks the scale of the operating asset base. For the IT-consulting company, the primary value drivers (talent, client relationships, proprietary processes) are internally generated and expensed as incurred rather than capitalised, so book value captures only a small fraction of the company's true economic asset base.

D. Final Result: P/B is meaningful for the cement producer but poorly suited to the IT-consulting company.

E. Interpretation: A high P/B multiple for the IT company does not necessarily indicate overvaluation — it may simply reflect that most of its true assets are invisible on the balance sheet; conversely, historical-cost depreciation can understate the cement producer's replacement value of plant and machinery in an inflationary environment, meaning even the 'more meaningful' P/B for the cement producer is not perfectly clean either.

F. Decision/Recommendation: The analyst should rely primarily on P/B (alongside other measures) for the cement producer, but should largely set aside P/B for the IT-consulting company in favour of earnings- or cash-flow-based multiples (P/E, EV/EBITDA), which better capture the value of its people-driven business model.

G. Teaching Insight: This case is a good opportunity to discuss how the usefulness of any single valuation multiple depends on the underlying accounting treatment of a company's key assets — encourage students to ask 'what does book value actually capture here?' before applying P/B mechanically across sectors.

SECTION VII

P/S and P/CF Valuation

Case 46: Stable-Growth Justified P/S

Learning Objective: Derive a justified P/S from net profit margin, payout, growth, and required return.

Case Situation

A listed food-processing company maintains a stable net profit margin of 8% and a dividend payout ratio of 50%. Its return on equity is 16% and its cost of equity is 13%. The company's sales per share is ₹150. An analyst wants to estimate the justified P/S multiple and infer a value per share from it.

Required

113. Estimate the sustainable growth rate and the justified P/S multiple.
114. Estimate the value per share implied by the justified P/S and sales per share.

Solution

A. Relevant Valuation Principle: The justified P/S multiple links net margin, payout, sustainable growth, and the required return, allowing analysts to value companies directly from sales when earnings-based multiples are less reliable.

B. Formula: g = b × ROE; Justified P/S = Net Margin × Payout × (1+g) / (Ke − g); Value per share = Justified P/S × Sales per share

C. Substitution and Working:

b = 1 − 0.50 = 0.50; g = 0.50 × 0.16 = 8.0%

Justified P/S = 0.08 × 0.50 × 1.08 / (0.13 − 0.08) = 0.0432/0.05 = 0.864x

Value per share = 0.864 × 150 = ₹129.60

D. Final Result: Justified P/S ≈ 0.864x; implied value per share ≈ ₹129.60.

E. Interpretation: The justified P/S multiple ties directly to the company's net margin and reinvestment economics — a higher margin or higher sustainable growth (within reason) would mechanically raise the justified multiple, since P/S is essentially a margin-adjusted version of the same underlying DDM logic used for P/E and P/B.

F. Decision/Recommendation: The ₹129.60 value-per-share estimate can be compared against the current market price for a straightforward Buy/Hold/Avoid recommendation, keeping in mind that P/S-based value is highly sensitive to the assumed net margin.

G. Teaching Insight: A common student error is omitting the (1+g) term in the numerator, or confusing net margin with operating (EBIT) margin. Reinforce that the justified-P/S formula is the same underlying DDM structure, just expressed per rupee of sales instead of per rupee of earnings.

Case 47A: Two-Stage P/S Valuation — Pharmacy Chain

Learning Objective: Estimate P/S when sales grow rapidly and payout changes after the high-growth period.

Case Situation

An organised pharmacy chain currently has sales per share of ₹100 and expects sales growth of 25% annually for the next five years as it expands into new cities. During this expansion, net margin is thin at 4% and payout is low at 10%, as most cash is reinvested. Thereafter, sales growth stabilises at 8%, with net margin improving to 6% and payout rising to 50%. The cost of equity is 14%. An analyst wants the two-stage justified P/S and a comparison with the current market multiple.

Required

1. Project sales per share and dividends per share for each of the five high-growth years.
2. Estimate the stable-stage terminal value and the resulting two-stage justified P/S.
3. Compare the result with the market multiple, assuming the stock currently trades at 1.2x sales.

Solution

A. Relevant Valuation Principle: A two-stage P/S valuation separately values the thin-margin, low-payout high-growth-phase dividends and a stable-stage terminal value based on improved margin and payout, then expresses the result as a multiple of current sales.

B. Formula: Sales_t = Sales_(t-1) × (1+g_hg); D_t = Sales_t × margin_hg × payout_hg; TV5 = D6/(Ke − g_stable); Implied P/S = V0/Sales0

C. Substitution and Working:

Sales (₹, years 1–5): 125.00, 156.25, 195.31, 244.14, 305.18

Dividends (4% margin × 10% payout): ₹0.50, 0.62, 0.78, 0.98, 1.22

PV of each dividend (at 14%): ₹0.44, 0.48, 0.53, 0.58, 0.63

PV of high-growth-phase dividends = ₹2.66

Sales6 = 305.18×1.08 = ₹329.59; D6 = 329.59×0.06×0.50 = ₹9.89
TV5 = 9.89/(0.14−0.08) = ₹164.79; PV of TV5 = ₹85.59
V0 = 2.66+85.59 = ₹88.25; Implied P/S = 88.25/100 = 0.88x

D. Final Result: Implied two-stage justified P/S ≈ 0.88x, against a market multiple of 1.2x.

E. Interpretation: Despite the pharmacy chain's rapid 25% sales growth, the thin margins and low payout during the expansion phase mean very little value comes from the explicit high-growth dividends — almost all the implied value (about 97%) comes from the terminal value once margins and payout improve. The market's 1.2x multiple sits meaningfully above the 0.88x justified figure, suggesting the market expects either faster margin improvement, a longer growth runway, or lower risk than assumed here.

F. Decision/Recommendation: Given the market multiple exceeds the fundamentals-based justified P/S, the analyst should treat the stock as fully valued to slightly expensive at 1.2x sales, and should specifically probe the margin-improvement assumption before considering a Buy.

G. Teaching Insight: This case shows students that rapid revenue growth alone does not justify a high P/S multiple if margins remain thin during the growth phase — the terminal-stage margin assumption is doing almost all of the work in this valuation, similar to the terminal-value dominance seen in DDM and DCF cases.

Case 47B: Two-Stage P/S Valuation — Quick-Service Restaurant Chain

Learning Objective: Estimate P/S when sales grow very rapidly off a thin-margin, low-payout base, and recognize when a market multiple implies an unsustainable growth or margin story.

Case Situation
Zaika Quick Service Restaurants Ltd, an Indian QSR chain, currently has sales per share of ₹80 and expects aggressive sales growth of 30% annually for the next four years as it opens new outlets across tier-2 cities. During this expansion, net margin is thin at 3% and payout is minimal at 5%, since almost all cash is reinvested into store build-out. Thereafter, sales growth moderates to 7%, with net margin improving to 7% and payout rising to 40%. The cost of equity is 13%. An analyst wants the two-stage justified P/S and a comparison with the current market multiple.

Required

1. Project sales per share and dividends per share for each of the four high-growth years.
2. Estimate the stable-stage terminal value and the resulting two-stage justified P/S.
3. Compare the result with the market multiple, assuming the stock currently trades at 1.5x sales.

Solution

A. Relevant Valuation Principle: A two-stage P/S valuation separately values the thin-margin, low-payout high-growth-phase dividends and a stable-stage terminal value based on improved margin and payout, then expresses the result as a multiple of current sales.

B. Formula: Sales_t = Sales_(t-1) × (1+g_hg); D_t = Sales_t × margin_hg × payout_hg; TV4 = D5/(Ke – g_stable); Implied P/S = V0/Sales0

C. Substitution and Working:
Sales (₹, years 1–4): 104.00, 135.20, 175.76, 228.49
Dividends (3% margin × 5% payout): ₹0.16, 0.20, 0.26, 0.34
PV of each dividend (at 13%): ₹0.14, 0.16, 0.18, 0.21
PV of high-growth-phase dividends = ₹0.69
Sales5 = 228.49×1.07 = ₹244.48; D5 = 244.48×0.07×0.40 = ₹6.85
TV4 = 6.85/(0.13−0.07) = ₹114.09; PV of TV4 = ₹69.97
V0 = 0.69+69.97 = ₹70.66; Implied P/S = 70.66/80 = 0.88x

D. Final Result: Implied two-stage justified P/S ≈ 0.88x, against a market multiple of 1.5x.

E. Interpretation: Despite a punishing 30% sales growth rate, the terminal value contributes about 99% of total value — the explicit four-year growth phase, at a 3% margin and 5% payout, contributes almost nothing. The market's 1.5x multiple sits far above the 0.88x justified figure — a much wider gap than a typical growth story would suggest, implying the market is pricing in either a materially longer runway of 30% growth, a much faster margin ramp than modelled, or is simply chasing the growth headline without discounting it.

F. Decision/Recommendation: The gap here (roughly 70% above the justified multiple) is too large to attribute to reasonable optimism alone. The analyst should treat the stock as significantly overvalued at 1.5x sales and should specifically challenge whether store-level unit economics can plausibly improve margin from 3% to 7% as quickly as assumed.

G. Teaching Insight: This case is a deliberately more extreme version of the terminal-value-dominance lesson: even a very high headline growth rate is nearly worthless to value if it comes at a thin margin and minimal payout. Students should notice that the size of the market-justified gap matters — a 36% gap (as in a typical case) invites a "fully valued" conclusion, while a 70% gap like this one should prompt a harder challenge to the market's implicit assumptions, not just a mild caution.

Case 47C: Two-Stage P/S Valuation — Specialty Packaged Foods Company

Learning Objective: Estimate P/S for a moderate-growth company and recognize that a two-stage justified multiple can also reveal undervaluation, not just overvaluation.

Case Situation
Priyam Specialty Foods Ltd, an Indian branded packaged-foods company, currently has sales per share of ₹150 and expects steady sales growth of 20% annually for the next six years as its products gain shelf space in modern retail. During this expansion, net margin is 5% and payout is 15%. Thereafter, sales growth settles at 6%, with net margin improving to 8% and payout rising to 60%. The cost of equity is 12%. An analyst wants the two-stage justified P/S and a comparison with the current market multiple.

Required

1. Project sales per share and dividends per share for each of the six high-growth years.
2. Estimate the stable-stage terminal value and the resulting two-stage justified P/S.
3. Compare the result with the market multiple, assuming the stock currently trades at 1.0x sales.

Solution

A. Relevant Valuation Principle: A two-stage P/S valuation separately values the moderate-margin, moderate-payout high-growth-phase dividends and a stable-stage terminal value based on improved margin and payout, then expresses the result as a multiple of current sales.

B. Formula: Sales_t = Sales_(t-1) × (1+g_hg); D_t = Sales_t × margin_hg × payout_hg; TV6 = D7/(Ke − g_stable); Implied P/S = V0/Sales0

C. Substitution and Working:
Sales (₹, years 1–6): 180.00, 216.00, 259.20, 311.04, 373.25, 447.90
Dividends (5% margin × 15% payout): ₹1.35, 1.62, 1.94, 2.33, 2.80, 3.36
PV of each dividend (at 12%): ₹1.21, 1.29, 1.38, 1.48, 1.59, 1.70
PV of high-growth-phase dividends = ₹8.65
Sales7 = 447.90×1.06 = ₹474.77; D7 = 474.77×0.08×0.60 = ₹22.79
TV6 = 22.79/(0.12−0.06) = ₹379.82; PV of TV6 = ₹192.43
V0 = 8.65+192.43 = ₹201.08; Implied P/S = 201.08/150 = 1.34x

D. Final Result: Implied two-stage justified P/S ≈ 1.34x, against a market multiple of 1.0x.

E. Interpretation: Here the relationship reverses: the market's 1.0x multiple sits below the fundamentals-based justified figure of 1.34x. With a moderate 5% margin, a meaningful 15% payout even during the growth phase, and a credible improvement to an 8% margin thereafter, the terminal value (about 96% of total value) is well supported by realistic assumptions — yet the market is pricing the stock as if growth or margin expansion will disappoint.

F. Decision/Recommendation: Unlike Cases 47A and 47B, this case supports a Buy: the stock appears undervalued by roughly 25% relative to its justified multiple, provided the analyst is confident the 8% stable-stage margin is achievable — a materially smaller assumption stretch than the margin jump required in Case 47B.

G. Teaching Insight: Place this case immediately after 47A and 47B to make the point explicit: the two-stage P/S method is not a one-directional "growth stocks are overvalued" tool — it is equally capable of flagging undervaluation. What changes the verdict is not the growth rate itself, but the credibility of the margin and payout assumptions supporting the terminal value in each case.

Case 48: Margin and P/S Interpretation Matrix

Learning Objective: Interpret P/S jointly with profitability rather than treating a low multiple as automatically attractive.

Case Situation

Four Indian consumer companies are being screened by a research desk: Company W has a low P/S of 0.8x and a low net margin of 3%; Company X has a low P/S of 0.9x but a high net margin of 12%; Company Y has a high P/S of 3.5x and a high net margin of 15%; and Company Z has a high P/S of 3.2x but a low net margin of 2%. A junior analyst initially assumes the two low-P/S companies (W and X) are automatically the more attractive investments.

Required

115. Classify each company as potentially undervalued, overvalued, or reasonably valued based on the joint P/S-margin relationship.
116. Explain why a low P/S multiple alone should not be treated as automatically attractive.

Solution

A. Relevant Valuation Principle: P/S must be interpreted jointly with net margin: a low P/S combined with a low margin may simply reflect a genuinely low-quality, low-profitability business (fairly valued, not cheap), while a low P/S combined with a high margin is a stronger signal of genuine undervaluation; the reverse logic applies to high P/S multiples.

B. Formula: (Conceptual case — classification based on the joint P/S and margin combination, not a single calculation.)

C. Substitution and Working:

Company W (low P/S 0.8x, low margin 3%): the low multiple is likely justified by weak profitability — reasonably valued, not necessarily cheap.

Company X (low P/S 0.9x, high margin 12%): a low multiple despite strong profitability is a genuine red flag for potential undervaluation, worth further investigation.

Company Y (high P/S 3.5x, high margin 15%): the high multiple is at least partly justified by strong profitability — plausibly reasonably valued, though further growth-durability analysis is needed.

Company Z (high P/S 3.2x, low margin 2%): a high multiple despite weak profitability is a red flag for potential overvaluation.

D. Final Result: Classification: Company W — reasonably valued; Company X — potentially undervalued; Company Y — plausibly reasonably valued; Company Z — potentially overvalued.

E. Interpretation: The instinct to treat both low-P/S companies (W and X) as equally attractive is incorrect: Company W's low multiple is consistent with, and likely explained by, its weak margin, while Company X's low multiple stands out precisely because it is not explained by profitability — making X, not W, the more interesting undervaluation candidate.

F. Decision/Recommendation: The research desk should prioritise deeper due diligence on Company X (low P/S, high margin) as the strongest candidate for genuine undervaluation, and should treat Company Z (high P/S, low margin) as the strongest candidate for overvaluation, while W and Y warrant only routine monitoring.

G. Teaching Insight: This 2x2 margin-and-multiple matrix is a useful teaching device for showing that valuation multiples only become meaningful once paired with the profitability metric that would justify them — a lesson applicable well beyond P/S to virtually every relative-valuation multiple.

Case 49: P/S for a Loss-Making Digital Business

Learning Objective: Use P/S cautiously when earnings are negative but revenue and unit economics are observable.

Case Situation

A loss-making online marketplace has sales per share of ₹40 and is growing revenue rapidly, with an improving path toward profitability as unit economics (contribution per order) turn positive. Comparable listed and recently listed marketplace businesses trade at P/S multiples ranging from 2.0x to 3.5x, with the higher end typically reserved for businesses further along their margin-improvement path. An analyst must estimate an indicative value despite the absence of positive earnings.

Required

117. Estimate an indicative value per share using an appropriately chosen P/S multiple from the comparable range.
118. Identify why current revenue quality and future margin conversion are critical to the reliability of this estimate.

Solution

A. Relevant Valuation Principle: P/S is one of the few multiples usable for loss-making businesses, but its reliability depends heavily on whether current revenue is high-quality (sustainable, not incentive-driven) and whether the path to positive margins is credible — the multiple itself says nothing about profitability.

B. Formula: Indicative Value per Share = Selected P/S Multiple × Sales per Share

C. Substitution and Working:

Given the company's improving but not yet proven margin path, a multiple toward the lower-middle of the comparable range (2.5x) is more appropriate than the top of the range: Indicative value = 2.5 × 40 = ₹100.00

D. Final Result: Indicative value per share ≈ ₹100.00, using a 2.5x P/S multiple.

E. Interpretation: This estimate is only as reliable as the assumption that current revenue is sustainable (not a product of heavy discounting or promotional spend that will unwind) and that the company's contribution margin will continue improving as suggested; if either assumption fails, the appropriate P/S multiple — and hence the indicative value — could be considerably lower.

F. Decision/Recommendation: The analyst should present ₹100.00 as an indicative, not firm, value, explicitly flagging revenue quality and margin-conversion risk as the two key assumptions underlying the choice of a mid-range rather than top-of-range multiple.

G. Teaching Insight: This case reinforces that P/S is a workaround for the absence of earnings, not a substitute for scrutinising whether the underlying revenue and unit economics are genuinely improving — students should resist treating any P/S-based number for a loss-making business as more certain than it is.

Case 50: P/CF and Earnings Quality

Learning Objective: Compare earnings-based and cash-flow-based multiples when accruals are significant.

Case Situation

Two listed engineering companies, Company M and Company N, both have an EPS of ₹10.00 and trade at the same market price of ₹150, giving both an identical P/E of 15x. However, Company M's operating cash flow per share is only ₹8.00, reflecting a build-up in receivables and inventory, while Company N's operating cash flow per share is ₹14.00, reflecting much better collection and inventory discipline. An analyst wants to look beyond the identical P/E to assess which company deserves a higher valuation.

Required

119. Calculate the P/CF ratio for both companies.
120. Identify which company shows better cash conversion.
121. Explain why the company with better cash conversion may deserve a higher valuation despite the identical P/E.

Solution

A. Relevant Valuation Principle: P/CF, based on operating cash flow rather than accrual earnings, can reveal meaningful differences in earnings quality between companies that appear identically valued on a P/E basis.

B. Formula: P/CF = Market Price per Share / Operating Cash Flow per Share

C. Substitution and Working:

Company M: P/CF = 150/8.00 = 18.75x

Company N: P/CF = 150/14.00 = 10.71x

D. Final Result: P/CF: Company M = 18.75x; Company N = 10.71x, despite both having an identical P/E of 15x.

E. Interpretation: Company N converts a much larger share of its reported earnings into actual cash (CFO/EPS of 1.4x, versus 0.8x for Company M), indicating higher earnings quality and lower risk that reported profit is inflated by unconverted receivables or inventory build-up. On a cash-flow basis, Company N is meaningfully cheaper (10.71x versus 18.75x) despite trading at the same P/E as Company M.

F. Decision/Recommendation: Given its superior cash conversion, Company N arguably deserves a higher valuation (or Company M a discount) relative to the identical P/E the market currently assigns both companies — an analyst constructing a relative-value view should favour Company N on this basis, all else equal.

G. Teaching Insight: This case is a strong illustration that two companies with an identical P/E are not necessarily equally attractively valued — P/CF is a useful cross-check specifically when working-capital dynamics differ meaningfully between otherwise-similar companies.

SECTION VIII

Enterprise Value, Non-Operating Assets, Beta, and APV

Case 51: Enterprise Value Bridge

Learning Objective: Move correctly from market capitalisation to enterprise value.

Case Situation

An Indian packaging company has an equity market value of ₹800 crore. It carries borrowings of ₹300 crore, cash and equivalents of ₹60 crore, preference share capital of ₹40 crore, and minority interest of ₹25 crore relating to a partly-owned subsidiary. An analyst preparing a comparable-company analysis needs the enterprise value, which must include all claims on the business beyond common equity.

Required

122. Calculate the enterprise value.
123. Identify which claims must be included in, or excluded from, the enterprise value bridge, and why.

Solution

A. Relevant Valuation Principle: Enterprise value represents the value of the entire operating business, so it must include all financial claims other than common equity (debt, preference capital, minority interest) and exclude cash, which is not part of core operations.

B. Formula: Enterprise Value = Market Capitalisation + Borrowings + Preference Capital + Minority Interest − Cash

C. Substitution and Working:

EV = 800 + 300 + 40 + 25 − 60 = ₹1,105 crore

D. Final Result: Enterprise value = ₹1,105 crore.

E. Interpretation: Borrowings, preference capital, and minority interest all represent claims on the operating business that sit alongside, or ahead of, common equity — so they must be added to market capitalisation, not ignored; cash, by contrast, is a non-operating asset that reduces the amount an acquirer would effectively need to pay for the operating business, so it is subtracted.

F. Decision/Recommendation: The ₹1,105 crore enterprise value, not the ₹800 crore equity market value, should be used as the numerator for any EV-based multiple (such as EV/EBITDA) in the comparable-company analysis.

G. Teaching Insight: A common student error is omitting minority interest and preference capital, treating debt and cash as the only two adjustment items. Reinforce that any claim senior to or alongside common equity belongs in the bridge, regardless of how small it appears.

Case 52: Valuing Excess Cash and Marketable Securities

Learning Objective: Separate cash required for operations from genuinely excess cash.

Case Situation

A mature IT-services company holds ₹500 crore in cash and short-term investments on its balance sheet. Based on the company's working-capital cycle and typical operating needs, a reasonable estimate of the minimum operating cash required to run the business smoothly is ₹100 crore. The company's core operating-asset (enterprise) value has been separately estimated at ₹3,000 crore, and it carries no debt. An analyst must decide how much of the ₹500 crore balance-sheet cash should be treated as genuinely excess and added to arrive at equity value.

Required

124. Calculate the excess cash available to be added to operating-asset value.
125. Estimate the resulting equity value.
126. Explain why all balance-sheet cash should not automatically be treated as surplus.

Solution

A. Relevant Valuation Principle: Only cash beyond what is needed to run day-to-day operations should be treated as excess and added to enterprise value when deriving equity value; the portion needed for operating liquidity is already implicitly required to sustain the operating-asset value itself.

B. Formula: Excess Cash = Total Cash and Investments − Minimum Operating Cash Required; Equity Value = Enterprise Value + Excess Cash (no debt in this case)

C. Substitution and Working:

Excess cash = 500 − 100 = ₹400 crore

Equity value = 3,000 + 400 = ₹3,400 crore

D. Final Result: Excess cash = ₹400 crore; equity value ≈ ₹3,400 crore.

E. Interpretation: Treating the full ₹500 crore as excess cash would overstate equity value by ₹100 crore, since that portion is necessary working capital that supports the day-to-day operations already reflected in the ₹3,000 crore enterprise value — double-counting it would inflate the valuation.

F. Decision/Recommendation: The analyst should use ₹3,400 crore, not ₹3,500 crore, as the estimated equity value, and should document the ₹100 crore minimum operating cash assumption clearly for review.

G. Teaching Insight: A frequent error is treating 100% of reported cash as 'excess' simply because a company appears to hold more cash than seems necessary; reinforce that a reasonable minimum operating cash estimate — even if approximate — must always be netted out first.

Case 53: Cross-Holdings and Subsidiary Investments

Learning Objective: Incorporate the value of investments in associates and subsidiaries without double counting.

Case Situation

A listed Indian holding company owns a 15% stake in separately listed Company A (market capitalisation ₹2,000 crore) and a 20% stake in separately listed Company B (market capitalisation ₹1,500 crore). Its own standalone operating business is separately valued at ₹1,200 crore. Given the typically poor marketability and governance discount associated with holding-company structures in India, the analyst has been instructed to apply a 25% holding-company discount to the value of the cross-holdings before adding them to standalone operating value.

Required

127. Calculate the pre-discount value of the two cross-holdings.
128. Apply the holding-company discount and estimate the total equity value.
129. Explain why the discount, not the full market value of the stakes, should be added.

Solution

A. Relevant Valuation Principle: The value of a holding company's cross-holdings must be based on its proportional stake in each investee's market value, adjusted by an appropriate holding-company discount that reflects illiquidity, governance concerns, and the market's typical scepticism toward conglomerate structures — added to, not blended with, standalone operating value.

B. Formula: Cross-holding Value = Σ (Stake % × Investee Market Cap); Adjusted Value = Cross-holding Value × (1 − Discount); Total Equity Value = Standalone Operating Value + Adjusted Cross-holding Value

C. Substitution and Working:

Stake in A = 0.15 × 2,000 = ₹300 crore

Stake in B = 0.20 × 1,500 = ₹300 crore

Total pre-discount cross-holding value = 300+300 = ₹600 crore

Adjusted value (after 25% discount) = 600 × 0.75 = ₹450 crore

Total equity value = 1,200 + 450 = ₹1,650 crore

D. Final Result: Total equity value ≈ ₹1,650 crore, comprising ₹1,200 crore standalone value and ₹450 crore of discounted cross-holdings.

E. Interpretation: Without the holding-company discount, the cross-holdings would be valued at their full ₹600 crore market value, but Indian holding-company structures typically trade at a discount to the sum of their parts because minority shareholders in the holding company cannot directly access the underlying investee cash flows or dividends without leakage through the holding structure.

F. Decision/Recommendation: The analyst should use ₹1,650 crore, not the undiscounted ₹1,800 crore, as the total equity value, and should be prepared to justify the specific 25% discount against observed market discounts for comparable Indian holding structures.

G. Teaching Insight: A common error is adding the full market value of cross-holdings without any discount, ignoring the well-documented empirical tendency for holding companies to trade below the sum of their parts. Encourage students to treat the discount rate itself as a judgment call requiring justification, not a fixed constant.

Case 54: Surplus Land and Idle Assets

Learning Objective: Value non-operating real estate separately from the operating business.

Case Situation

A century-old textile company owns a large parcel of unused urban land, originally acquired decades ago for a mill that has since been shut down. The land is carried on the books at its historical cost of ₹5 crore, but current market estimates value it at ₹150 crore given the surrounding area's commercial development. The applicable long-term capital gains tax rate on the sale of this land is 20%. An analyst valuing the company's equity must decide how to incorporate this asset without simply using either the book value or the full quoted market value.

Required

130. Estimate the after-tax realisable value of the land.
131. Explain why neither the book value nor the full market value should be added to operating value without adjustment.

Solution

A. Relevant Valuation Principle: Non-operating real estate should be valued at its realistic, after-tax realisable value — reflecting both the capital-gains tax that would arise on an eventual sale and, where relevant, the uncertainty and time involved in actually monetising the asset — not at its low historical book value or its unadjusted headline market value.

B. Formula: After-tax Realisable Value = Market Value − Tax on Capital Gain, where Capital Gain = Market Value − Book Value

C. Substitution and Working:

Capital gain = 150 − 5 = ₹145 crore

Tax on gain = 145 × 0.20 = ₹29 crore

After-tax realisable value = 150 − 29 = ₹121 crore

D. Final Result: After-tax realisable value of the land ≈ ₹121 crore.

E. Interpretation: Using the book value of ₹5 crore would drastically understate the asset's true worth, while using the full ₹150 crore market value would overstate it by ignoring the tax that would be triggered on an eventual sale; ₹121 crore is a more defensible estimate, though a further conservative haircut may be warranted given the typical uncertainty and time lag involved in actually monetising large urban land parcels.

F. Decision/Recommendation: The analyst should add approximately ₹121 crore (potentially with an additional modest liquidity/uncertainty discount, at the analyst's judgment) to the company's operating-asset value, clearly documenting both the tax adjustment and the rationale for any further haircut applied.

G. Teaching Insight: This case reinforces that non-operating asset valuation requires the same rigour as operating valuation — book value is frequently irrelevant for old, appreciated real estate, and the 'market value' quoted in press reports or broker estimates is rarely a clean, realisable number without tax and liquidity adjustments.

Case 55: Unlevering the Beta

Learning Objective: Remove the effect of financial leverage from an observed equity beta.

Case Situation

An NSE-listed auto-parts company, being used as the closest available comparable for valuing an unlisted target in the same industry, has an observed equity beta of 1.30. The company's debt-equity ratio is 0.60, and the applicable corporate tax rate is 25%. Before this beta can be used for the unlisted target, which has a different capital structure, an analyst must first strip out the effect of the comparable company's own financial leverage.

Required

132. Calculate the comparable company's asset (unlevered) beta.
133. Interpret what this asset beta represents as a measure of operating risk.

Solution

A. Relevant Valuation Principle: Unlevering a comparable company's equity beta removes the amplifying effect of its specific financial leverage, isolating the asset beta that reflects only the operating (business) risk common to the industry — a necessary first step before relevering to a different target capital structure.

B. Formula: Beta(asset) = Beta(equity) / [1 + (1 − t) × (D/E)]

C. Substitution and Working:

Beta(asset) = 1.30 / [1 + (0.75 × 0.60)] = 1.30 / [1 + 0.45] = 1.30/1.45 = 0.90

D. Final Result: Asset (unlevered) beta ≈ 0.90.

E. Interpretation: An asset beta of 0.90, below the observed equity beta of 1.30, indicates that a meaningful portion of the comparable company's equity risk (as measured by its levered beta) comes from financial leverage rather than from the underlying operating risk of the auto-parts business itself; 0.90 represents the operating-risk measure common to unlevered auto-parts businesses generally.

F. Decision/Recommendation: This asset beta of 0.90 is the correct figure to carry forward and relever using the unlisted target's own capital structure, rather than applying the comparable company's raw 1.30 equity beta directly.

G. Teaching Insight: A common student error is applying a comparable company's equity beta directly to a target with a different capital structure, without first unlevering and relevering. Reinforce that beta comparisons across companies are only valid at the asset (unlevered) level.

Case 56: Relevering the Beta for a Target Capital Structure

Learning Objective: Convert a comparable company's asset beta into an equity beta suitable for the target company.

Case Situation

Continuing from the auto-parts comparable analysis, the unlisted manufacturing target being valued is expected to operate at a higher debt-equity ratio of 1.0, compared with the comparable company's 0.60. Using the previously derived asset beta of 0.90 and the same 25% tax rate, an analyst must relever the beta to reflect the target's own capital structure, then estimate its cost of equity given a risk-free rate of 7% and a market-risk premium of 6%.

Required

134. Relever the beta for the target's debt-equity ratio of 1.0.
135. Estimate the target's cost of equity using the relevered beta.
136. Explain the effect of the higher leverage on the cost of equity, compared with the original comparable company's beta.

Solution

A. Relevant Valuation Principle: Relevering applies a target's own debt-equity ratio to the (industry-common) asset beta, producing an equity beta — and hence a cost of equity — appropriate to the target's specific capital structure, which may differ meaningfully from that of the original comparable company.

B. Formula: Beta(target equity) = Beta(asset) × [1 + (1 − t) × (D/E)_target]; Ke = Rf + Beta × Market Risk Premium

C. Substitution and Working:

Beta(target) = 0.90 × [1 + (0.75 × 1.0)] = 0.90 × 1.75 = 1.57

Ke(target) = 0.07 + 1.57 × 0.06 = 0.07 + 0.0942 = 16.42%

(For comparison, Ke using the original comparable's beta of 1.30 would be: 0.07 + 1.30×0.06 = 14.80%)

D. Final Result: Relevered target beta ≈ 1.57; target cost of equity ≈ 16.42%, versus 14.80% implied by the comparable company's own (higher-leverage-inappropriate) beta.

E. Interpretation: Because the target plans to operate at a higher debt-equity ratio (1.0 versus the comparable's 0.60), its equity holders bear more financial risk per unit of operating risk, correctly raising both the relevered beta and the cost of equity relative to what the comparable company's raw beta would suggest.

F. Decision/Recommendation: The analyst should use the target-specific cost of equity of approximately 16.42%, not the comparable company's implied 14.80%, in any FCFE-based or WACC-based valuation of the target.

G. Teaching Insight: This case, paired with Case 55, completes the full unlever-relever workflow — emphasise that skipping either step (using a raw comparable beta, or unlevering without relevering to the target's actual structure) produces a materially wrong cost of equity.

Case 57: Adjusted Present Value under Changing Debt

Learning Objective: Apply APV when financing effects can be valued separately from operating assets.

Case Situation

An infrastructure project company has an unlevered (all-equity) business value estimated at ₹500 crore. The project is financed with a clearly scheduled declining debt balance of ₹300 crore, ₹250 crore, ₹200 crore, ₹150 crore, and ₹100 crore over five years, at an interest rate of 10% and a corporate tax rate of 25%. Rather than estimate a single blended WACC over a debt schedule that changes every year, the analyst has chosen to apply the adjusted present value (APV) approach, valuing the interest tax shields separately.

Required

137. Calculate the annual interest tax shield for each of the five years.
138. Calculate the present value of the interest tax shields.
139. Estimate the APV and explain why this approach can be cleaner than a single WACC under changing leverage.

Solution

A. Relevant Valuation Principle: APV values the unlevered business separately from the present value of financing side-effects (primarily interest tax shields), avoiding the need to estimate a single, constant WACC when the capital structure changes materially and predictably over the forecast period.

B. Formula: Tax Shield_t = Debt_t × Interest Rate × Tax Rate; PV(Tax Shields) = Σ Tax Shield_t × $(1+Kd)^{-t}$; APV = Unlevered Value + PV(Tax Shields)

C. Substitution and Working:

Tax shields (₹ crore): Year1 = 300×0.10×0.25=7.50; Year2=250×0.025=6.25; Year3=200×0.025=5.00; Year4=150×0.025=3.75; Year5=100×0.025=2.50

PV of tax shields (discounted at Kd = 10%) ≈ ₹19.85 crore

APV = 500 + 19.85 = ₹519.85 crore

D. Final Result: PV of interest tax shields ≈ ₹19.85 crore; APV ≈ ₹519.85 crore.

E. Interpretation: Because the debt schedule declines predictably and specifically over five years, APV allows the analyst to value the unlevered project on its own operating merits (₹500 crore) and add precisely the tax-shield benefit this particular financing schedule provides (₹19.85 crore), rather than trying to average a constantly changing debt-equity mix into a single WACC figure that would be technically inconsistent with a materially varying capital structure.

F. Decision/Recommendation: The analyst should report an equity/project value based on the APV of ₹519.85 crore, and should be able to show the tax-shield component separately if debt levels or interest-rate assumptions are later revised.

G. Teaching Insight: This case highlights APV's key practical advantage — it isolates financing effects into a separately auditable component (the tax shields), which is particularly valuable for infrastructure and project-finance valuations where scheduled debt paydowns are common and a single WACC would obscure the year-by-year financing benefit.

SECTION IX

EVA, MVA, and Value Creation

Case 58: Calculating Economic Value Added

Learning Objective: Determine whether operating profit is sufficient to cover the cost of invested capital.

Case Situation

An Indian engineering company reported NOPAT (net operating profit after tax) of ₹90 crore for the year. The company's total invested capital — the sum of debt and equity capital deployed in the business — is ₹600 crore, and its WACC is estimated at 12%. The CFO wants to know whether the company's operating profit was sufficient to cover the cost of the capital employed to generate it, beyond simply reporting a positive net profit.

Required

140. Calculate the capital charge for the year.
141. Calculate EVA for the year.
142. State whether the company created or destroyed economic value.

Solution

A. Relevant Valuation Principle: EVA measures whether NOPAT exceeds the dollar (rupee) cost of the capital invested to generate it — a positive accounting profit does not automatically mean economic value has been created.

B. Formula: Capital Charge = Invested Capital × WACC; EVA = NOPAT − Capital Charge

C. Substitution and Working:

Capital charge = 600 × 0.12 = ₹72 crore

EVA = 90 − 72 = ₹18 crore

D. Final Result: Capital charge = ₹72 crore; EVA = ₹18 crore.

E. Interpretation: With EVA of ₹18 crore, the company earned meaningfully more than the cost of the capital deployed in the business during the year — indicating genuine economic value creation, not merely accounting profitability.

F. Decision/Recommendation: The company can be assessed as having created shareholder value during the year; this positive EVA supports continued or expanded investment in the business, all else equal.

G. Teaching Insight: Reinforce to students that EVA is not simply 'profit' — it explicitly charges the business for the capital tied up in generating that profit, which is precisely what distinguishes it from standard accounting measures like net profit or even NOPAT alone.

Case 59: Comparing Accounting Profit and EVA

Learning Objective: Understand why a profitable company can still destroy shareholder value.

Case Situation

A public-sector manufacturing company reported a substantial net profit after tax of ₹150 crore for the year, which management has highlighted as a strong result. However, the company employs a very large capital base of ₹2,000 crore, reflecting decades of accumulated investment in plant and equipment, and its WACC is estimated at 10%. A board member questions whether the headline profit figure alone tells the full story about value creation.

Required

143. Calculate the capital charge for the year.
144. Calculate EVA for the year.
145. Explain why a positive net profit does not guarantee value creation.

Solution

A. Relevant Valuation Principle: A large, profitable-looking company can still destroy economic value if its capital base is so large that the absolute cost of capital exceeds the absolute profit generated — profitability and value creation are related but distinct concepts.

B. Formula: Capital Charge = Invested Capital × WACC; EVA = NOPAT − Capital Charge

C. Substitution and Working:

Capital charge = 2,000 × 0.10 = ₹200 crore

EVA = 150 − 200 = −₹50 crore

D. Final Result: Capital charge = ₹200 crore; EVA = −₹50 crore.

E. Interpretation: Despite reporting a healthy ₹150 crore net profit, the company's very large ₹2,000 crore capital base means the cost of capital employed (₹200 crore) exceeds the profit generated — the company is economically destroying value even while remaining accounting-profitable, likely because the invested capital is not being deployed efficiently enough relative to its scale.

F. Decision/Recommendation: The board should not treat the ₹150 crore net profit as evidence of good capital allocation; management should be pressed to explain the low return on this very large capital base and to identify opportunities to either improve capital efficiency or reduce underutilised capital.

G. Teaching Insight: This case directly confronts the common misconception that any positive net profit reflects value creation. It is an excellent anchor case for explaining why EVA-based performance measurement is used by sophisticated boards and investors alongside, not instead of, standard profit metrics.

Case 60: Market Value Added

Learning Objective: Measure the difference between the market value supplied by investors and the capital invested in the business.

Case Situation

A listed Indian consumer company has a market value of equity of ₹1,800 crore and a market value of debt of ₹200 crore. The total capital originally invested in the business (book capital employed, reflecting historical equity and debt contributions) is ₹1,200 crore. A shareholder wants to understand what the difference between the company's total market value and its invested capital reveals about the company's track record of value creation.

Required

146. Calculate the total market value of the company.
147. Calculate the market value added (MVA).
148. Interpret what a positive MVA figure indicates.

Solution

A. Relevant Valuation Principle: MVA measures the cumulative difference between what the market believes the business is worth today and the capital originally invested in it — a positive MVA reflects investors' expectation that the company has generated, and will continue to generate, returns above its cost of capital.

B. Formula: Total Market Value = Market Value of Equity + Market Value of Debt; MVA = Total Market Value − Invested Capital

C. Substitution and Working:

Total market value = 1,800 + 200 = ₹2,000 crore

MVA = 2,000 − 1,200 = ₹800 crore

D. Final Result: Total market value = ₹2,000 crore; MVA = ₹800 crore.

E. Interpretation: A positive MVA of ₹800 crore indicates that the market values the company's ongoing operations at ₹800 crore more than the capital originally invested to build it — reflecting the market's collective assessment that the company has generated, and is expected to continue generating, positive EVA over time. MVA can be thought of as the present value of expected future EVA.

F. Decision/Recommendation: The shareholder can view the positive MVA as supportive evidence of a track record of value-creating capital allocation, though it should be considered alongside a direct EVA trend analysis rather than in isolation.

G. Teaching Insight: MVA is best introduced immediately after EVA to reinforce the link between them: MVA is essentially the market's capitalised view of expected future EVA. A common error is treating MVA as unrelated to EVA rather than as its market-based counterpart.

Case 61: Linking EVA to Business Decisions and MVA

Learning Objective: Evaluate whether a growth project will improve economic value and potentially support market value.

Case Situation

A company is considering an expansion into a new Indian state, requiring additional invested capital of ₹150 crore. The expansion is expected to generate incremental NOPAT of ₹25 crore per year once fully operational, and the company's WACC is 13%. Management wants to know whether this specific expansion decision is financially value-creating before committing capital, and how such a decision might relate to the company's future market value added.

Required

149. Calculate the incremental capital charge and incremental EVA associated with the expansion.
150. Recommend whether the expansion is financially value-creating.
151. Explain how this decision relates to the company's future MVA, while noting the additional factors MVA depends on.

Solution

A. Relevant Valuation Principle: A specific investment decision can be evaluated on a standalone basis by comparing its incremental NOPAT against its incremental capital charge; if incremental EVA is positive, the project is expected to add to the company's cumulative economic value creation, which over time should be reflected in market value added, provided the market shares this expectation.

B. Formula: Incremental Capital Charge = Additional Invested Capital × WACC; Incremental EVA = Incremental NOPAT − Incremental Capital Charge

C. Substitution and Working:

Incremental capital charge = 150 × 0.13 = ₹19.50 crore

Incremental EVA = 25 − 19.50 = ₹5.50 crore

D. Final Result: Incremental capital charge = ₹19.50 crore; incremental EVA = ₹5.50 crore (positive).

E. Interpretation: The positive incremental EVA of ₹5.50 crore indicates the expansion is expected to earn more than its cost of capital, making it financially value-creating on a standalone basis, though this conclusion depends on the NOPAT and capital estimates proving reasonably accurate once the expansion is operational.

F. Decision/Recommendation: Management should proceed with the expansion on financial grounds, since it is expected to be value-creating; however, MVA will only rise if the broader market comes to share this expectation of sustained positive EVA — MVA depends on the market's overall view of the company's future prospects, not solely on this one project's realised or expected EVA.

G. Teaching Insight: This case closes the EVA-MVA loop by connecting a single capital-budgeting decision to the broader concept of market value added, while cautioning students that a single value-creating project does not mechanically translate into a proportionate rise in market value, since MVA reflects the market's aggregate view of the entire business's future.

SECTION X

LBO and Integrated Valuation Applications

Case 62: LBO Sources, Uses, and Debt Capacity

Learning Objective: Understand the basic structure of a leveraged buyout transaction.

Case Situation

A domestic private equity fund is considering the acquisition of a mid-sized packaging company at a purchase enterprise value of ₹900 crore. Transaction expenses (legal, advisory, and financing fees) are estimated at ₹20 crore. The financing plan includes senior debt of ₹400 crore, subordinated debt of ₹150 crore, and a management rollover (existing management reinvesting part of their proceeds as equity) of ₹50 crore. The fund needs a simple sources-and-uses summary before finalising the deal structure.

Required

152. Prepare a sources-and-uses summary for the transaction.
153. Estimate the sponsor's (private equity fund's) required equity contribution.

Solution

A. Relevant Valuation Principle: An LBO's sources-and-uses summary must balance: total uses (purchase price plus transaction costs) must exactly equal total sources (debt, management rollover, and sponsor equity) — the sponsor's equity contribution is the balancing figure after all debt and rollover sources are accounted for.

B. Formula: Total Uses = Purchase Enterprise Value + Transaction Expenses; Sponsor Equity = Total Uses − (Senior Debt + Subordinated Debt + Management Rollover)

C. Substitution and Working:

Total uses = 900 + 20 = ₹920 crore

Debt + rollover = 400 + 150 + 50 = ₹600 crore

Sponsor equity = 920 − 600 = ₹320 crore

D. Final Result: Total uses = ₹920 crore; sponsor equity required ≈ ₹320 crore.

E. Interpretation: The sponsor is funding roughly 35% of the transaction (₹320 crore of ₹920 crore) with its own equity, while debt (senior and subordinated combined) and management rollover fund the remaining 65% — a fairly typical leverage level for a mid-sized Indian LBO, though the appropriate mix depends on the target's cash-flow stability and debt-servicing capacity.

F. Decision/Recommendation: The fund can proceed with this structure provided the target's cash flows (assessed in a subsequent debt-servicing analysis) can comfortably support the combined senior and subordinated debt of ₹550 crore.

G. Teaching Insight: This case establishes the basic sources-and-uses framework that underlies every LBO; a common student error is forgetting to include transaction expenses in total uses, or forgetting management rollover as a genuine source (not equivalent to sponsor equity) when solving for the sponsor's contribution.

Case 63: Cash Sweep and Debt Repayment

Learning Objective: Analyse how operating cash flow reduces acquisition debt and increases sponsor equity value.

Case Situation

Following its acquisition (continuing from the packaging-sector LBO context), an auto-component company begins with an opening acquisition debt balance of ₹550 crore, carrying an average interest rate of 10%. Over a three-year forecast, EBITDA is expected at ₹180 crore, ₹198 crore, and ₹216 crore; depreciation at ₹30 crore, ₹32 crore, and ₹34 crore; capital expenditure at ₹40 crore, ₹35 crore, and ₹32 crore; and working-capital investment at ₹10 crore, ₹8 crore, and ₹6 crore, for years 1, 2, and 3 respectively. The tax rate is 25%. All available cash after interest, tax, capex, and working capital is swept to repay debt each year.

Required

154. Estimate the cash available for debt repayment in each of the three years.
155. Estimate the closing debt balance at the end of each year.

Solution

A. Relevant Valuation Principle: A cash sweep applies all free cash generated after interest, tax, and reinvestment needs directly toward debt repayment, causing the debt balance — and therefore the following year's interest expense — to decline progressively faster than a simple fixed amortisation schedule would.

B. Formula: Cash available for debt repayment = EBITDA − Interest − Tax − Capex − ΔWC, where Tax = (EBITDA − Depreciation − Interest) × t; Closing Debt = Opening Debt − Cash Available for Repayment

C. Substitution and Working:

Year 1: Interest = 550×10% = ₹55.0 cr; EBIT = 180−30 = ₹150 cr; Tax = (150−55)×0.25 = ₹23.75 cr; Cash available = 180−55−23.75−40−10 = ₹51.25 cr; Closing debt = 550−51.25 = ₹498.75 cr

Year 2: Interest = 498.75×10% = ₹49.875 cr; EBIT = 198−32 = ₹166 cr; Tax = (166−49.875)×0.25 = ₹29.03 cr; Cash available = 198−49.875−29.03−35−8 = ₹76.09 cr; Closing debt = 498.75−76.09 = ₹422.66 cr

Year 3: Interest = 422.66×10% = ₹42.27 cr; EBIT = 216−34 = ₹182 cr; Tax = (182−42.27)×0.25 = ₹34.93 cr; Cash available = 216−42.27−34.93−32−6 = ₹100.80 cr; Closing debt = 422.66−100.80 = ₹321.86 cr

D. Final Result: Cash available for debt repayment: ₹51.25 crore, ₹76.09 crore, ₹100.80 crore (years 1–3). Closing debt: ₹498.75 crore, ₹422.66 crore, ₹321.86 crore.

E. Interpretation: The cash sweep is self-reinforcing: as debt falls each year, interest expense declines, freeing up more cash for repayment the following year — evident in the accelerating debt paydown, from ₹51.25 crore repaid in year 1 to ₹100.80 crore repaid in year 3, even before accounting for the underlying EBITDA growth.

F. Decision/Recommendation: By the end of year 3, acquisition debt has fallen from ₹550 crore to approximately ₹321.86 crore — a reduction of over 41% — which directly and substantially increases the equity value available to the sponsor at any future exit, independent of any change in the business's enterprise value.

G. Teaching Insight: This case demonstrates the core economic engine of an LBO: debt paydown mechanically transfers enterprise value to equity holders over the holding period, even with no change in the underlying business value — a distinct source of sponsor returns separate from EBITDA growth or multiple expansion.

Case 64: Exit Value and Sponsor IRR

Learning Objective: Estimate private equity returns from entry price, debt reduction, exit multiple, and holding period.

Case Situation

A private equity fund invested ₹300 crore of equity to acquire an Indian healthcare-services company and plans to exit after a five-year holding period. At exit, the company's EBITDA is expected to be ₹250 crore, and comparable healthcare-services transactions suggest an exit multiple of 9.0x EV/EBITDA. Acquisition debt is expected to have been paid down to ₹400 crore remaining at the time of exit. The fund's investment committee wants the expected money multiple and approximate IRR on the original equity investment.

Required

156. Calculate the exit enterprise value and exit equity value.
157. Calculate the money multiple on the original sponsor equity investment.
158. Estimate the approximate IRR over the five-year holding period.

Solution

A. Relevant Valuation Principle: Private equity returns are driven by the combination of EBITDA growth, debt paydown, and any change in valuation multiple between entry and exit; the money multiple and IRR translate the resulting exit equity value back into a return on the original equity invested.

B. Formula: Exit EV = Exit EBITDA × Exit Multiple; Exit Equity Value = Exit EV − Remaining Debt; Money Multiple = Exit Equity Value / Initial Sponsor Equity; IRR ≈ (Money Multiple)^(1/Holding Period) − 1

C. Substitution and Working:

Exit EV = 250 × 9.0 = ₹2,250 crore

Exit equity value = 2,250 − 400 = ₹1,850 crore

Money multiple = 1,850/300 = 6.17x

IRR ≈ (6.17)^(1/5) − 1 ≈ 43.88%

D. Final Result: Exit equity value ≈ ₹1,850 crore; money multiple ≈ 6.17x; approximate IRR ≈ 43.88% over the five-year holding period.

E. Interpretation: A money multiple above 6x and an IRR well above typical private-equity hurdle rates reflect the combined effect of strong EBITDA growth (implied, given the exit EBITDA level), substantial debt paydown over the holding period, and a healthy exit multiple — an outcome that should be cross-checked against how much of the return is attributable to each driver before being treated as a base-case expectation.

F. Decision/Recommendation: The investment committee should decompose this return into its EBITDA-growth, debt-paydown, and multiple components before approving the underlying operating assumptions as the base case, since an IRR this strong warrants scrutiny of whether the 9.0x exit multiple and expected debt paydown are realistic rather than optimistic.

G. Teaching Insight: This case is a good closing point for the LBO sequence, showing how the sources-and-uses (Case 62) and cash-sweep (Case 63) mechanics combine with market-based exit assumptions to produce the return metrics (money multiple, IRR) that ultimately matter to a private equity investor.

Case 65: Integrated Equity Valuation and Recommendation

Learning Objective: Reconcile DCF and relative valuation outputs into a defensible investment conclusion.

Case Situation

A fictional NSE-listed consumer-electronics company, Ujjwal Electronics Ltd, has been valued by an equity research team using three separate methods: an FCFF-based DCF value of ₹320 per share, a justified P/E-based value of ₹300 per share, and a peer EV/EBITDA-based value of ₹340 per share. The three methods produce a reasonable but not identical valuation range, and the team lead has asked for a single reconciled target price and a clear investment recommendation, along with the principal risk to the thesis.

Required

159. Assign reasonable weights to the three valuation methods and compute a weighted target price.
160. Identify the principal downside risk to the recommendation.
161. Provide a concise investment recommendation.

Solution

A. Relevant Valuation Principle: When multiple valuation methods produce a reasonable but non-identical range, a defensible target price is typically constructed by weighting each method according to its reliability for the specific company and business context, rather than by mechanically averaging or arbitrarily picking the highest or lowest figure.

B. Formula: Weighted Target Price = Σ (Weight_i × Value_i)

C. Substitution and Working:
Assigning 40% weight to the FCFF-based DCF value (reflecting its direct link to company-specific cash-flow fundamentals), 30% to the justified P/E value, and 30% to the peer EV/EBITDA value:

Weighted target = 0.40×320 + 0.30×300 + 0.30×340 = 128 + 90 + 102 = ₹320.00

D. Final Result: Weighted target price ≈ ₹320.00 per share.

E. Interpretation: The reconciled target of ₹320.00 sits close to the DCF-based value, reflecting the higher weight assigned to it as the method most directly grounded in the company's own cash-flow fundamentals, while still incorporating the market's relative-valuation perspective through the P/E and EV/EBITDA components; the principal downside risk is a demand slowdown or margin compression in the consumer-electronics category, which would simultaneously lower the FCFF forecast, the justified P/E (via lower sustainable ROE), and the peer EV/EBITDA multiple (via sector-wide de-rating) — meaning the three methods are not fully independent checks against this specific risk.

F. Decision/Recommendation: A Buy recommendation is supported if the current market price sits meaningfully below ₹320.00, with the explicit caveat that a category-wide demand slowdown or margin compression is the key risk that could simultaneously undermine all three valuation approaches rather than being diversified away by using multiple methods.

G. Teaching Insight: This integrative case is best used to show that using three valuation methods does not automatically triple the robustness of a conclusion if all three share a common vulnerability to the same underlying risk factor (in this case, sector demand and margins) — true diversification of valuation risk requires methods sensitive to different, not the same, underlying drivers.

SECTION XI

Strategic Acquisitions, Synergies, and Negotiation

Case 66: Debt Capacity from an Interest-Coverage Constraint

Learning Objective: Size acquisition debt from a lender's minimum interest-coverage requirement, distinct from a target debt-to-value ratio.

Case Situation

A private equity-backed strategic acquirer is evaluating a leveraged acquisition of a regional dairy-products company with trailing EBITDA of ₹240 crore. The lending syndicate has indicated it will finance the deal only if annual interest is covered at least 4.5 times by EBITDA, and has priced the debt at 9.5%. The acquirer's own separately derived estimate of the target's enterprise value is ₹1,800 crore, and it wants to know how much acquisition debt this coverage constraint actually supports before finalising a financing structure.

Required

162. Calculate the maximum annual interest the target's EBITDA can support under the lender's coverage constraint.
163. Calculate the implied acquisition debt capacity.
164. Express this debt capacity as a percentage of the independently estimated enterprise value, and comment on whether this confirms the debt is affordable.

Solution

A. Relevant Valuation Principle: Lenders in an acquisition financing typically size debt off a minimum interest-coverage ratio applied to the target's own EBITDA, rather than an arbitrary debt-to-value target; this produces a defensible ceiling on the interest burden the business can support at the point of acquisition.
B. Formula: Maximum Interest = EBITDA / Coverage Ratio; Debt Capacity = Maximum Interest / Cost of Debt
C. Substitution and Working:
Maximum interest = 240 / 4.5 = ₹53.33 crore.
Debt capacity = 53.33 / 0.095 = ₹561.40 crore.
D. Final Result: Debt capacity ≈ ₹561.40 crore, supporting a maximum annual interest of ₹53.33 crore.
E. Interpretation: The implied debt of ₹561.40 crore represents approximately 31.2% of the independently estimated ₹1,800 crore enterprise value — a moderate, seemingly conservative leverage level by acquisition-finance standards.
F. Decision/Recommendation: This figure establishes only the initial financing ceiling permitted by the coverage covenant; it says nothing about whether the target's actual free cash flow can service both interest and the scheduled repayment of this principal over time, which must be tested separately.
G. Teaching Insight: A frequent student error is to treat debt capacity, once sized from a coverage ratio, as automatically serviceable. Coverage measures only the interest burden; it does not test whether cash flow after tax, capital expenditure, and working-capital investment is sufficient to also repay principal — the next case in this section builds directly on this distinction.

Case 67: The Bullet Repayment and the Coverage-versus-Cash-Sufficiency Distinction

Learning Objective: Distinguish debt capacity (an interest-coverage test) from debt-service feasibility (a full cash-flow test including scheduled principal and a bullet repayment).

Case Situation

Continuing the dairy-products acquisition from the previous case, the ₹560 crore acquisition debt is structured to amortise monthly as if over eight years, but carries a bullet repayment at the end of year 4 equal to whatever principal remains outstanding at that point. Scheduled annual interest and principal for years 1–4 are: Year 1 — interest ₹51.10 crore, principal ₹49.10 crore; Year 2 — interest ₹46.23 crore, principal ₹53.98 crore; Year 3 — interest ₹40.87 crore, principal ₹59.33 crore; Year 4 — interest ₹34.98 crore, principal (bullet) ₹397.59 crore. Projected unlevered free cash flow for the same four years is ₹70 crore, ₹78 crore, ₹85 crore, and ₹90 crore. The tax rate is 25%.

Required

165. Calculate after-tax interest and total debt service (after-tax interest plus scheduled principal) for each of the four years.
166. Calculate annual excess cash (unlevered free cash flow minus total debt service) for each year.
167. Comment on what this reveals about the financing structure that the interest-coverage test in the previous case did not.

Solution

A. Relevant Valuation Principle: A complete debt-service test must subtract both after-tax interest and scheduled principal — including any bullet repayment — from the target's own unlevered free cash flow; passing an interest-coverage covenant at the point of acquisition says nothing about whether cash flow can fund the full repayment schedule.
B. Formula: After-Tax Interest = Interest × (1 − Tax Rate); Total Debt Service = After-Tax Interest + Scheduled Principal; Excess Cash = Unlevered Free Cash Flow − Total Debt Service
C. Substitution and Working:
Year 1: after-tax interest = 51.10×0.75 = 38.33; total service = 38.33+49.10 = 87.43; excess = 70−87.43 = −17.43.
Year 2: after-tax interest = 46.23×0.75 = 34.67; total service = 34.67+53.98 = 88.65; excess = 78−88.65 = −10.65.
Year 3: after-tax interest = 40.87×0.75 = 30.65; total service = 30.65+59.33 = 89.98; excess = 85−89.98 = −4.98.
Year 4: after-tax interest = 34.98×0.75 = 26.24; total service = 26.24+397.59 = 423.83; excess = 90−423.83 = −333.83.
D. Final Result: Excess cash is negative in every year: −₹17.43 crore, −₹10.65 crore, −₹4.98 crore, and −₹333.83 crore in years 1 through 4 respectively.
E. Interpretation: Despite comfortably clearing the lender's 4.5× interest-coverage covenant at the point of acquisition (Case 66), the target's own free cash flow cannot fund scheduled principal in any of the four years, and the shortfall becomes severe once the year-4 bullet falls due — a lump-sum repayment nearly 4.4 times the target's entire fourth-year free cash flow.
F. Decision/Recommendation: The acquirer must arrange committed refinancing, a revolving credit facility, or an equity contribution well before year 4, and should revisit whether the original financing structure — sized purely off interest coverage — is actually appropriate for this business's cash-generation profile.
G. Teaching Insight: This case, read together with the previous one, makes the central point of acquisition-debt analysis concrete: an interest-coverage ratio and a cash-sufficiency test are different questions, and a financing structure can pass the first while failing the second — precisely because coverage ignores principal repayment altogether.

Case 68: Un-levering and Re-levering Under Two Different Capital-Structure Assumptions

Learning Objective: Apply the correct beta formula for two different capital-structure assumptions — a comparable's current, unchanging debt level versus a target's own permanent long-run debt-to-value ratio — and show that the two formulas are not interchangeable.

Case Situation

An analyst has already derived an unlevered (asset) beta of 0.75 for a logistics-sector comparable set, and the applicable tax rate is 30%. The analyst must now relever this beta for two distinct purposes: first, an acquisition-day capital structure with a debt-to-equity ratio of 1.2 that is expected to change materially as acquisition debt is paid down; and second, a permanent long-run target capital structure corresponding to an industry-average debt-to-value ratio of 30%, expected to hold constant indefinitely once the business stabilises.

Required

168. Relever the beta for the acquisition-day debt-to-equity ratio of 1.2, using the formula appropriate for a changing capital structure.
169. Relever the beta for the permanent target debt-to-value ratio of 30%, using the formula appropriate for a constant capital structure, and calculate the corresponding cost of equity (risk-free rate 7%, market risk premium 6%).
170. Recompute part 2 using the same formula as part 1 instead, and explain why the two approaches give different answers even though the underlying business risk has not changed.

Solution

A. Relevant Valuation Principle: Relevering a beta for a capital structure expected to keep changing calls for the constant-debt formula (the same formula used to un-lever a comparable's beta); relevering for a capital structure assumed to remain permanently constant calls for a different, constant-D/V formula. Applying the wrong one of the two produces a beta, and therefore a cost of equity, that does not correctly reflect the assumption actually being made.
B. Formula: Constant-debt relevering: Beta(target) = Beta(asset) × [1 + (1−t)×(D/E)]. Constant-D/V relevering: Beta(target) = Beta(asset) / (1 − D/V)
C. Substitution and Working:
(1) Acquisition-day (constant-debt formula, D/E=1.2): Beta = 0.75 × [1 + 0.70×1.2] = 0.75 × 1.84 = 1.380.
(2) Permanent target (constant-D/V formula, D/V=30%): Beta = 0.75 / (1−0.30) = 0.75/0.70 = 1.071; Ke = 0.07 + 1.071×0.06 = 13.43%.
(3) The same D/V of 30% implies a D/E of 0.30/0.70 = 0.4286. Using the constant-debt formula instead: Beta = 0.75 × [1+0.70×0.4286] = 0.75×1.30 = 0.975; Ke = 0.07+0.975×0.06 = 12.85%.
D. Final Result: Correct permanent-structure beta ≈ 1.071 (Ke ≈ 13.43%); if the constant-debt formula is mistakenly applied at the same D/E instead, beta ≈ 0.975 (Ke ≈ 12.85%) — a difference of nearly 60 basis points in the cost of equity from using the wrong formula alone.
E. Interpretation: Both calculations start from the identical 0.75 asset beta and reach the identical 30% target D/V, yet produce different equity betas, because they encode different assumptions about how that leverage behaves over time — a constantly rebalanced ratio (constant-D/V) versus a fixed debt amount that mechanically changes the ratio as firm value moves (constant-debt). The two are not interchangeable approximations of the same thing.
F. Decision/Recommendation: The analyst should use the acquisition-day relevered beta (1.380) only for the near-term period while acquisition debt is being paid down on a known schedule, and the permanent-structure beta (1.071) only for the terminal, steady-state period — never one formula for both.
G. Teaching Insight: This is one of the most commonly confused steps in leveraged valuation. A useful classroom check: ask why the same formula used to un-lever a comparable's current beta cannot also be used to relever a target's permanent target beta — the answer is that un-levering a comparable's

observed beta and relevering for an acquisition-day structure both assume a fixed debt amount, while relevering for a stable long-run target assumes a fixed ratio instead.

Case 69: Combining APV and WACC in One Going-Concern Valuation

Learning Objective: Value a going-concern business using APV for an explicit period with a known, changing debt schedule, and WACC for the terminal period once leverage is assumed to stabilise — combined into a single valuation.

Case Situation

An acquirer is valuing a target with a four-year explicit forecast period, over which acquisition debt is paid down on a known schedule, followed by an indefinite terminal period during which leverage is assumed to settle at a permanent, industry-normal level. Unlevered free cash flow for the four explicit years is ₹60 crore, ₹66 crore, ₹72 crore, and ₹78 crore, discounted at an unlevered cost of capital of 11%. Annual interest expense over the same four years is ₹24 crore, ₹20 crore, ₹16 crore, and ₹12 crore, with tax shields valued at a 30% tax rate and discounted at the 10% cost of debt. Beyond year 4, cash flow is expected to grow at a stable 5% in perpetuity, discounted at a terminal WACC of 10.5%.

Required

171. Calculate the present value of the explicit-period unlevered free cash flow and the present value of the interest tax shields.
172. Calculate the present value of the terminal value, using the terminal WACC and long-run growth rate.
173. Combine these into a total valuation, and explain why using a single discount rate throughout — either the unlevered cost of capital or the terminal WACC — would be inappropriate.

Solution

A. Relevant Valuation Principle: When a target's capital structure is expected to change on a known schedule during an explicit forecast period and then stabilise permanently thereafter, the explicit period is valued using APV (unlevered cash flow plus separately valued tax shields), while the terminal period — where a constant capital structure is genuinely assumed — is valued using WACC; combining both methods in one valuation correctly matches each period's actual financing assumption.
B. Formula: PV(UFCF) = ΣUFCFt/(1+ra)^t; PV(Tax Shields) = Σ(Interesttxt)/(1+Kd)^t; TV(year4) = FCF4×(1+g)/(WACC−g); PV(TV) = TV(year4)/(1+WACC)^4; Total Value = PV(UFCF) + PV(Tax Shields) + PV(TV)
C. Substitution and Working:
PV(UFCF) at 11% = ₹211.65 crore.
Tax shields = ₹7.20cr, ₹6.00cr, ₹4.80cr, ₹3.60cr; PV at 10% = ₹17.57 crore.
Intermediate value = 211.65+17.57 = ₹229.22 crore.
TV(year4) = 78×1.05/(0.105−0.05) = ₹1,489.09 crore; PV(TV) at 10.5% over 4 years = ₹998.79 crore.
D. Final Result: Total value = 229.22 + 998.79 = ₹1,228.00 crore.
E. Interpretation: Roughly 81% of total value comes from the terminal period alone, underscoring how heavily this valuation depends on the assumption that the target genuinely settles into a stable, permanent capital structure and growth rate beyond year 4 — the entire justification for switching to WACC at that point.
F. Decision/Recommendation: The acquirer should present both components of value separately — the APV-based explicit-period value and the WACC-based terminal value — rather than as a single blended number, so that the sensitivity of the (much larger) terminal component to the WACC and growth assumptions is visible to decision-makers.
G. Teaching Insight: A common error is to select one method and apply it throughout the entire valuation horizon. Using WACC during the explicit period is technically inconsistent because the capital

structure is changing on a known schedule (exactly what Case 57 already illustrates in isolation); using APV throughout the terminal period is unnecessarily cumbersome because the debt balance is no longer explicitly known, only a constant ratio — this case shows why a single valuation legitimately uses both.

Case 70: Enterprise-Level Sustainable Growth — Reinvestment Rate and Return on Capital

Learning Objective: Derive a firm-level (capital-based) sustainable growth rate for use in an FCFF or APV valuation, and contrast it with the equity-based g = b × ROE growth rate used elsewhere in this casebook.

Case Situation

A specialty-chemicals manufacturer's final forecast-year figures are: NOPAT ₹96 crore, capital expenditure ₹40 crore, increase in working capital ₹8 crore, and depreciation ₹30 crore, on a total invested capital base of ₹640 crore. Separately, the company's return on equity is 18% and its dividend payout ratio is 55%. An analyst preparing a terminal value for an FCFF-based valuation must decide which growth rate is appropriate for this purpose.

Required

174. Calculate the company's return on capital and its reinvestment rate.
175. Calculate the resulting firm-level (capital-based) sustainable growth rate.
176. Calculate what the growth rate would be using the equity-based g = b × ROE formula instead, and explain which figure is appropriate for an FCFF-based terminal value.

Solution

A. Relevant Valuation Principle: A terminal growth rate used in an FCFF or APV valuation should be derived from the same capital base that generates the firm-level cash flow being valued — return on capital and the associated reinvestment rate — rather than from an equity-level formula such as g = b × ROE, which reflects a different (and leverage-sensitive) base.
B. Formula: Return on Capital (ROC) = NOPAT / Invested Capital; Net Reinvestment = Capex + ΔWorking Capital − Depreciation; Reinvestment Rate = Net Reinvestment / NOPAT; g(capital-based) = Reinvestment Rate × ROC
C. Substitution and Working:
ROC = 96/640 = 15.00%.
Net reinvestment = 40+8−30 = ₹18 crore. Reinvestment rate = 18/96 = 18.75%.
g(capital-based) = 18.75% × 15.00% = 2.81%.
For comparison, g(equity-based) = b × ROE = (1−0.55) × 18% = 0.45 × 18% = 8.10%.
D. Final Result: Capital-based sustainable growth rate ≈ 2.81%; equity-based g = b × ROE would instead suggest ≈ 8.10% — nearly three times higher.
E. Interpretation: The two formulas diverge sharply here because ROE reflects the amplifying effect of financial leverage on shareholder returns, while ROC measures the return earned on the full capital base (debt and equity together) that actually funds the firm's operating reinvestment; an FCFF/APV valuation is built on unlevered, firm-level cash flow, so it must use the matching, firm-level growth driver.
F. Decision/Recommendation: The analyst should use 2.81%, not 8.10%, as the terminal growth rate in this FCFF-based valuation; using the equity-based figure here would materially overstate terminal value by embedding a growth rate the underlying operating capital base does not actually support.
G. Teaching Insight: Students who have only practised g = b × ROE (as in the dividend-discount and P/E, P/B, and P/S cases earlier in this casebook) commonly default to it here too. Reinforce that the growth formula must always match the cash-flow base being valued: equity cash flow and equity

returns for DDM/FCFE-style valuations, and capital-level cash flow and capital-level returns for FCFF/APV-style valuations.

Case 71: Valuing a Non-Controlling Affiliate Investment via an Earnings Multiple

Learning Objective: Value a non-controlling, unlisted equity-method investment separately from core operations, using an appropriate industry earnings multiple applied to the investor's share of the investee's profit, when the investee has no separately observable market value of its own.

Case Situation

A diversified industrial company holds a 30% non-controlling equity stake in an unlisted joint venture that reported net income of ₹40 crore for the year. Because the joint venture is not separately listed, no market capitalisation is directly observable for it, unlike the cross-holdings considered elsewhere in this casebook. A reasonable industry earnings multiple for comparable unlisted businesses of this type is 16.5x. The company's own standalone core operations have separately been valued at ₹2,100 crore.

Required

177. Calculate the investor's share of the joint venture's annual net income.
178. Apply the industry earnings multiple to estimate the value of this affiliate investment.
179. Calculate total equity value and comment on how this technique differs from valuing a cross-holding in a separately listed company.

Solution

A. Relevant Valuation Principle: When a non-controlling investment cannot be valued from its own observable market price because it is not separately listed, a reasonable approach is to apply an appropriate industry earnings multiple to the investor's proportionate share of the investee's reported profit, valuing the stake on a standalone earnings basis rather than folding its cash flows into the parent's own operating forecast.
B. Formula: Investor's Share of Income = Ownership % × Investee Net Income; Affiliate Value = Investor's Share of Income × Industry Earnings Multiple; Total Equity Value = Standalone Operating Value + Affiliate Value
C. Substitution and Working:
Investor's share of income = 0.30 × 40 = ₹12.00 crore.
Affiliate value = 12.00 × 16.5 = ₹198.00 crore.
Total equity value = 2,100 + 198.00 = ₹2,298.00 crore.
D. Final Result: Affiliate value ≈ ₹198.00 crore; total equity value ≈ ₹2,298.00 crore.
E. Interpretation: The affiliate investment contributes approximately 8.6% of total estimated value — material enough to warrant the same scrutiny of the assumed earnings multiple and reported profit figure as any other significant valuation input, even though it sits outside the company's core operating forecast.
F. Decision/Recommendation: This ₹198.00 crore should be reported and discussed as a distinct, non-operating component of value, clearly separated from the ₹2,100 crore core-operations value, so that the reliability of each can be assessed independently.
G. Teaching Insight: This case should be taught alongside the cross-holdings case earlier in this casebook, which values a stake in a separately listed investee at its own market capitalisation. Here, no such market value exists, so the analyst instead applies an earnings multiple to the investor's share of profit — a technique required whenever a non-controlling investment's own market value cannot be directly observed.

Case 72: Quantifying and Valuing Merger Synergies

Learning Objective: Quantify a cost synergy and a revenue synergy separately, convert each into after-tax incremental cash flow rather than stopping at a raw savings or revenue figure, and assess their relative credibility.

Case Situation

A consumer-products company is evaluating a merger expected to create two distinct sources of combination value. First, shared warehousing and distribution infrastructure is expected to reduce a combined distribution expense base — projected at ₹150 crore, ₹160 crore, and ₹170 crore over three years — by a phased 6%, 12%, and 18% respectively. Second, cross-selling to the merged entity's combined customer base is expected to add 1.5 lakh, 2.5 lakh, and 3.5 lakh new customers over the same three years, each spending approximately ₹8,000, ₹8,200, and ₹8,400 per year respectively, though management is far less certain about this customer-acquisition assumption than about the contractual distribution-cost savings. The tax rate is 30%, incremental revenue-synergy margin is assumed at 30% after costs, and both synergies are discounted at an unlevered cost of capital of 11%.

Required

180. Calculate the annual cost synergy and its after-tax present value.
181. Calculate the annual revenue synergy and its after-tax present value, applying the incremental margin assumption.
182. Compare the relative size and credibility of the two synergies, and comment on the implications for how much value should be attributed to each.

Solution

A. Relevant Valuation Principle: Merger synergies should be built up and valued separately by source — cost and revenue — because they typically differ sharply in evidentiary reliability; a raw expense saving or incremental revenue figure must also be converted to after-tax, margin-adjusted cash flow before being discounted, since revenue synergies in particular are not synonymous with cash-flow synergies.

B. Formula: Cost Synergyt = Base Expenset × Reduction %t; Revenue Synergyt = New Customerst × Average Spend per Customert; After-Tax Cash Flow = Synergyt × (1 − Margin/Cost Adjustment) × (1 − Tax Rate); PV = Σ After-Tax Cash Flowt / (1+r)^t

C. Substitution and Working:

Cost synergy: ₹9.00cr, ₹19.20cr, ₹30.60cr (years 1-3); after-tax: ₹6.30cr, ₹13.44cr, ₹21.42cr; PV at 11% = ₹32.25 crore.

Revenue synergy: ₹120.00cr, ₹205.00cr, ₹294.00cr; after 30% incremental margin and tax: ₹25.20cr, ₹43.05cr, ₹61.74cr; PV at 11% = ₹102.79 crore.

D. Final Result: PV of cost synergy ≈ ₹32.25 crore; PV of revenue synergy ≈ ₹102.79 crore; total synergy value ≈ ₹135.03 crore, of which the revenue synergy represents roughly 76% and the cost synergy roughly 24%.

E. Interpretation: The less certain of the two synergies — revenue, which depends on customer-acquisition assumptions management itself has flagged as uncertain — contributes more than three times the value of the more contractually grounded cost synergy; dollar magnitude and evidentiary credibility are pointing in opposite directions here, which should directly inform how much of this total the acquirer is willing to pay for.

F. Decision/Recommendation: The acquirer should treat the ₹135.03 crore as a ceiling estimate for negotiation purposes rather than a reliable base case, and should weight the cost-synergy component more heavily than the revenue-synergy component when deciding how much synergy value to actually price into a bid.

G. Teaching Insight: A common student error is to add up all identified synergies at face value without separately assessing each one's underlying reliability. Encourage students to always ask which synergies rest on assets or contracts the acquirer already controls (more credible) versus assumptions about customer or market behaviour that has not yet been demonstrated (less credible) — precisely the distinction this case is built to illustrate.

Case 73: From Synergy Value to a Negotiated Bid — Bargaining Range and Walk-Away Price

Learning Objective: Translate a standalone value (the floor) and a synergy-inclusive value (the ceiling) into a disciplined bargaining range and walk-away price, sharing synergy value with the seller rather than paying for all of it.

Case Situation

Continuing the merger evaluation from the previous case, the target's standalone enterprise value has independently been estimated at ₹850 crore, and the total present value of identified synergies is ₹135 crore, giving a synergy-inclusive enterprise value of ₹985 crore. The acquirer's negotiating team must decide how much of this ₹135 crore synergy value to offer the seller, recognising that paying for all of it would leave the acquirer no margin of safety if the synergies — particularly the less certain revenue component identified earlier — fail to fully materialise, and that an estimated ₹20.25 crore of integration costs (15% of synergy value) has not yet been deducted from any of these figures.

Required

183. Construct a bargaining-range table showing the implied bid at 0%, 25%, 50%, 75%, and 100% synergy-sharing with the seller.
184. Calculate a walk-away price using a 50% synergy-sharing anchor, net of the estimated integration-cost haircut.
185. Recommend where the opening bid should be set, and explain why paying the full synergy-inclusive value would be undisciplined.

Solution

A. Relevant Valuation Principle: The gap between a standalone value and a synergy-inclusive value is a bargaining range, not a single correct price; a disciplined acquirer shares synergy value with the seller only in proportion to the confidence and risk-bearing appropriate to each party, and separately deducts costs — such as integration costs — that the published synergy estimate has not yet reflected.
B. Formula: Implied Bid = Standalone EV + (Synergy Value × Share % Paid to Seller); Walk-Away Price = Implied Bid (at chosen share) − Integration-Cost Haircut
C. Substitution and Working:
At 0% share: 850 + 0 = ₹850.00cr. At 25%: 850 + 33.75 = ₹883.75cr. At 50%: 850 + 67.50 = ₹917.50cr. At 75%: 850 + 101.25 = ₹951.25cr. At 100%: 850+135 = ₹985.00cr.
Integration-cost haircut = 15% × 135 = ₹20.25cr.
Walk-away price (at 50% share) = 917.50 − 20.25 = ₹897.25cr.
D. Final Result: Bargaining range spans ₹850.00 crore (standalone floor) to ₹985.00 crore (full synergy ceiling); an illustrative walk-away price of approximately ₹897.25 crore emerges from a 50% synergy-sharing position net of estimated integration costs.
E. Interpretation: Anchoring the opening bid near the standalone floor and moving upward only as far as a defensible, risk-adjusted share of synergy value preserves a margin of safety for the acquirer if the revenue synergy identified in the previous case underperforms; paying close to the ₹985.00 crore ceiling would leave the acquirer with essentially no cushion and would implicitly assume the seller bears none of the execution risk.
F. Decision/Recommendation: The negotiating team should open below ₹897.25 crore, treat that figure as an approximate walk-away discipline rather than a fixed target, and be prepared to revisit

the synergy-sharing assumption explicitly if the seller produces credible evidence that improves confidence in the revenue-synergy component specifically.

G. Teaching Insight: This case closes the acquisition-valuation sequence in this casebook by making explicit what many students otherwise leave implicit: a calculated Enterprise Value, however carefully built, is not automatically a recommended purchase price — the final step from valuation to negotiation requires separate, deliberate judgment about value-sharing, unmodeled costs, and the discipline to walk away.

Summary of Principal Valuation Formulae

Valuation Gap %: Cases 1

Equity Value Bridge: Cases 2

Normalised PAT: Cases 3

Operating Cash Flow (Indirect Method): Cases 4

Free Cash Flow (CFO - Capex): Cases 5

CF/Discount-Rate Matching: Cases 6

Constant-Growth DDM: Cases 7, 17

Sustainable Growth (g=b×ROE): Cases 8

PVGO: Cases 9

Justified Value via g=b×ROE: Cases 10, 11

Two-Stage DDM: Cases 12

Transition-Period DDM: Cases 13

Three-Stage DDM: Cases 14

Multistage DDM from EPS/Payout: Cases 15

Naive Zero-Growth DDM: Cases 16

FCFF = EBIT(1-t)+Dep-Capex-ΔWC: Cases 18

FCFE = NI+Dep-Capex-ΔWC+NetBorrowing: Cases 19

FCFE=FCFF-Int(1-t)+NetBorrowing: Cases 20

FCFF & FCFE (levered vs unlevered): Cases 21

Explicit-Period FCFF Forecast: Cases 22

NOPAT & FCFF Forecast: Cases 23

FCFE with Net Debt Repayment: Cases 24

FCFF/WACC vs FCFE/Ke Selection: Cases 25

FCFE (Distributable, Regulatory Capital): Cases 26

EV via FCFF, Terminal Value, Equity Bridge: Cases 27

Normalised Terminal Cash Flow: Cases 28

Perpetuity-Growth Terminal Value: Cases 29

Guide for Selecting an Appropriate Valuation Method

1. Use the Dividend Discount Model (DDM) when a company has a stable, policy-driven dividend history that plausibly reflects its distributable earnings capacity.
2. Use FCFF/FCFE when dividends are unreliable, absent, or disconnected from earning capacity, or when a control/enterprise perspective is required.
3. Use FCFF (discounted at WACC) in preference to FCFE when the capital structure is expected to change materially over the forecast period.
4. Use FCFE directly for banks and NBFCs, where debt is integral to core operations rather than a separable financing choice.
5. Use justified P/E, P/B, or P/S when a quick, market-anchored cross-check is required, or when peer companies provide a useful relative benchmark.
6. Avoid P/E when earnings are negative; prefer P/S or a unit-economics measure in such cases.
7. Use EV/EBITDA-based exit multiples and APV for private equity, project-finance, and other contexts with clearly scheduled changes in leverage.
8. Use EVA and MVA to assess whether operating profit and market value reflect genuine economic value creation beyond the cost of capital employed.

Common Equity-Valuation Errors

1. Mismatching cash flow and discount rate (discounting FCFF at the cost of equity, or FCFE at WACC).
2. Using D0 instead of D1 in the Gordon growth formula.
3. Assuming a perpetual growth rate that exceeds, or is not meaningfully below, the discount rate or the long-term nominal GDP growth rate of the economy.
4. Treating any positive valuation gap as an automatic buy signal without assessing the reliability of the underlying assumptions.
5. Deducting gross debt without adding back cash, or ignoring minority interest and preference capital in the enterprise-value bridge.
6. Treating all balance-sheet cash as excess cash without netting out minimum operating cash requirements.
7. Applying a comparable company's raw equity beta to a target with a different capital structure without first unlevering and relevering it.
8. Treating a low P/E, P/B, or P/S multiple as automatically attractive without checking whether it is explained by correspondingly low profitability or growth.
9. Confusing accounting profit with economic value creation, ignoring the capital charge in EVA.
10. Treating a debt-funded increase in FCFE as evidence of operating value creation.

11. Treating debt capacity sized from an interest-coverage ratio as evidence that scheduled principal, including any bullet repayment, can also be serviced from operating cash flow.
12. Using the same beta un-levering formula to relever for both a temporary, changing capital structure and a permanent, constant long-run target structure.
13. Adding up identified merger synergies at face value without separately assessing the credibility of cost synergies versus revenue synergies, or without sharing synergy value with the seller in the final negotiated price.

Sensitivity-Analysis Guide

Because terminal value typically represents the majority of total enterprise or equity value in most DCF and DDM applications, every valuation should be presented with, at a minimum, a sensitivity range around the discount rate and the long-term growth assumption (see Cases 17 and 33 for worked illustrations). A useful practice is to recompute the valuation under one conservative and one optimistic combination of these two inputs, and to disclose whether the investment recommendation changes across that range.

In an acquisition context, sensitivity analysis should extend beyond the discount rate and growth assumption to the assumed synergy-realization levels and any integration-cost haircut, since these directly determine the bargaining range and walk-away price presented to a seller (see Cases 72 and 73).

Glossary of Important Valuation Terms

Intrinsic Value: An analyst's estimate of security's fundamental worth, typically based on the present value of expected future cash flows.

Enterprise Value: The value of a company's core operating assets, available to all providers of capital.

Equity Value: The value attributable to common shareholders, derived from enterprise value after adjusting for debt, cash, and other non-operating items.

Sustainable Growth Rate: The rate of growth a company can support through retained earnings, equal to the retention ratio multiplied by ROE.

Terminal Value: The value, at the end of an explicit forecast period, of all cash flows expected thereafter, usually estimated using a perpetuity-growth or exit-multiple approach.

Free Cash Flow to the Firm (FCFF): Cash flow available to all capital providers before financing effects, discounted at WACC.

Free Cash Flow to Equity (FCFE): Cash flow available specifically to equity shareholders after reinvestment and net debt movements, discounted at the cost of equity.

Justified Multiple: A valuation multiple (P/E, P/B, or P/S) derived from fundamentals such as payout, growth, ROE, and required return, used as a benchmark against the observed market multiple.

PVGO: The portion of a company's value attributable to expected future growth opportunities, as distinct from the value of assets already in place.

Economic Value Added (EVA): NOPAT less a charge for the cost of capital employed in the business; a measure of whether operating profit exceeds the cost of invested capital.

Market Value Added (MVA): The difference between a company's total market value and the capital originally invested in it.

Adjusted Present Value (APV): A valuation approach that separately values an unlevered business and the present value of financing side-effects, such as interest tax shields.

Interest Coverage Ratio: EBITDA divided by annual interest expense; the primary constraint lenders use to size acquisition debt capacity.
Bullet (Balloon) Repayment: A single lump-sum repayment of the entire remaining loan principal at a specified date, rather than the loan continuing to amortize on its original schedule.
Merger Synergy: Incremental value created specifically by combining two businesses, arising from cost savings, revenue growth, or both, and valued separately from each company's standalone worth.
Bargaining Range: The span between a target's standalone value (the floor) and its value inclusive of synergies (the ceiling), within which a negotiated acquisition price is expected to fall.
Walk-Away Price: The maximum price an acquirer commits, in advance of negotiation, not to exceed, reflecting a deliberate share of synergy value and a deduction for unmodeled integration or refinancing costs.

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